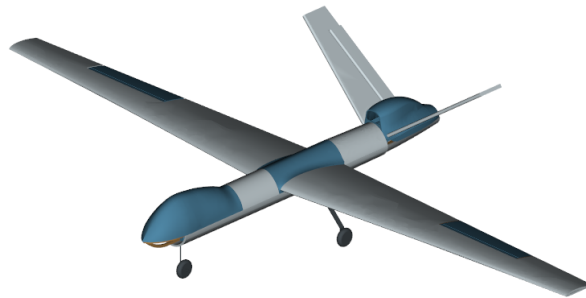


MQ9

Complete build guide

2 metre RC model / Revision 1 / 19 September 2026



Airframe fabrication, assembly, electronics and checks in one manual.

Cutting templates are separate under Manufacturing/Package/Print_Templates and inside the teammate ZIP. Print them at Actual Size /100%; the illustrations in this manual are not cutting templates.

This is a prototype fabrication handoff. Complete the material, dry-fit, structural, propulsion and flight checks described here. The provisional CG mark is not a flight-approved range.

Contents

03	Airframe manufacturing and assembly
15	Foam assembly map
20	Foam forming instructions
26	Electronics, calibration and CG
40	Detailed parts list
43	Hardware and forming notes
44	Plywood part atlas
48	Foam pattern overview
54	Plywood sheet overview
57	Foam sheet layouts
60	Motor mounting drawing
61	Verification and remaining physical checks

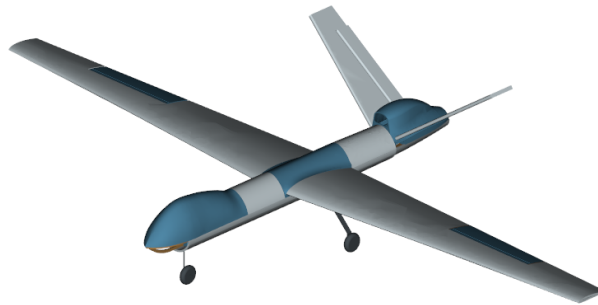
Who uses what

The teammate uses the airframe, foam, parts and hardware chapters. The owner uses the electronics and calibration chapter. Both complete the handoff and verification records.

Original sources remain under Manufacturing. Use the revised manufacturing files for declared changes; do not mix older packages with this release.

01 / MQ9 airframe

Manufacturing and assembly guide / 2 metre RC model / Revision 1



This guide is for the teammate fabricating and assembling the body. The owner fits and configures the electronics using the separate Electronics Build Guide. Use this revised package as a set; do not mix it with earlier AI conversion files.

Start here

1. Read this guide and the part registers. Print the fit coupons first.
2. Cut plywood and make the printed/metal parts. Mark every part ID.
3. Dry-assemble the complete skeleton and removable wings.
4. Form and fit one representative foam panel, then finish the skins.
5. Deliver the body with access open for electronics, plus the completed inspection record.

Release scope: dimensioned prototype fabrication files with declared corrections and digital checks. Material behaviour, landing loads, completed mass and flight performance have not been physically verified. The shop coupon, dry-fit and ground checks in this guide are required build operations, not claims that they have already passed.

02 / Files, tools and stock

Folder	Use
01_Plywood	Individual CUT profiles, quantity register and revised positioned STEP parts.
02_Foam	Developed knife patterns, forming guide, source mapping and seam evidence.
03_Print	Only selected aircraft print STLs, matching STEP and settings.
04_Hardware	Tube schedule, metal forming/drilling references, motor drawing and coupon.
05_Assembly	Revised assembly STEP and labelled reference illustrations.
06_Verification	Measured checks and declared revisions.
07_Source	Untouched original STEP assembly, individual parts ZIP and transcript.
Build Guide	Instructions, consolidated manual, part atlases and overviews.
Print_Templates	Actual-size A4 cutting and tooling templates. Keep these separate from the reading manual.

Machine work versus bench work

Laser or CNC: plywood CUT layer only. Do not cut labels, sanding guides, sheet edges or reference lines. Print: selected plastic parts. Metalwork: form/drill the gear references. Knife: the confirmed paper-faced XPS foam. Use a laser on XPS only with explicit workshop material/process approval.

Bench work remains essential: identify parts, check fit, bevel and form foam, glue, fit fasteners and linkages, align surfaces and inspect moving clearances. A machine export does not assemble the aircraft.

Stock and setup

Use aircraft/model plywood of the indicated thickness, including 2.5 mm nose doublers. Use the supplied 5 mm FliteBoard Pro, 1000 x 600 mm, nominal 400 g/m². Keep balsa optional. Use straight matched sliding carbon tubes as specified on page 6; 5 mm solid rods are not replacements.

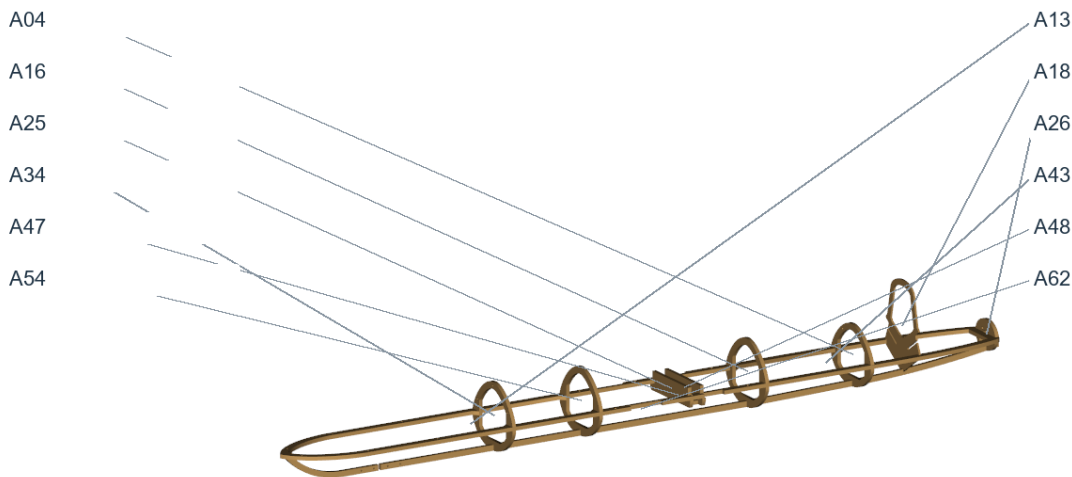
The reference plywood sheets are 1220 x 2440 mm; the occupied area is much smaller. Shops may reneat at 100% with quantities and grain directions preserved. Long spars/spine/doublers need face grain along their length. Never scale a part to fit a machine bed. Keep material, part IDs and sheet thickness together.

03 / Parts and buying list

Item	Quantity / specification
Plywood	31 cut pieces from 22 profiles: 24 nominal 5 mm, 4 at 3 mm, 3 at 2.5 mm. Two outer ribs are 4.990 mm CAD thickness; finish 5 mm stock to fit.
Foam	5 mm FliteBoard Pro. Follow per-pattern quantities; moving tail controls use 2 layers each. Ten supplied sheets are starting stock, not ten installed sheets.
Printed aircraft parts	11 individual STLs. One fit coupon is additional and is not installed.
Carbon	Front 8/6 mm joiner 406 mm x 1; rear 8/6 mm 390 mm x 1; matched 10 mm OD/8.2 mm nominal ID sleeves 199.35 mm x 2.
Wheels and gear	3 lightweight 50.8 mm wheels, 3 mm bores; formed main gear and 3 mm spring-wire nose gear; steering arm and shaft collars.
Wing root retention	6 M2 bolts, start 20 mm long; 12 washers and 6 locking nuts. Verify final stack.
Nose bypass	2 M2x 40-45 inner bolts +2 M2x 16 outer bolts; 8 washers, 4 locking nuts. Select actual length from stack.
Main gear fastening	2 M3x 16 through-bolts, large washers, 2 locking nuts; 2 M3 axle bolts, nominal 25 mm plus spacers/nuts.
Motor fastening	4 M3 screws, matching washers; determine screw length from 5 mm ply and actual safe motor thread depth.
Nose bearing	4 M3 bolts through bearing/mount, washers and locking nuts; start 16-20 mm and verify stack/clearance.
Controls	5 servos: 2 ailerons, 2 tail, 1 nose steering. 4 control horns, 1 steering arm, 5 pushrods with 10 clevises/ball links; servo screws.
Hinge pins	2 smooth 3 mm rods; 285 mm initial length, trim after fitting retainers.
Adhesives / covering	Wood-compatible epoxy/wood adhesive; XPS-compatible foam adhesive; reinforced tape and paper seam straps. Qualify on offcuts.

Stock fasteners are starting procurement lengths where the real nut, washer and servo horn affect the stack. Trim/select them at dry fit. Add spare washers, locking nuts, cable ties and servo extension leads. Do not use an unspecified screw through a motor base or into foam.

04 / Fuselage skeleton



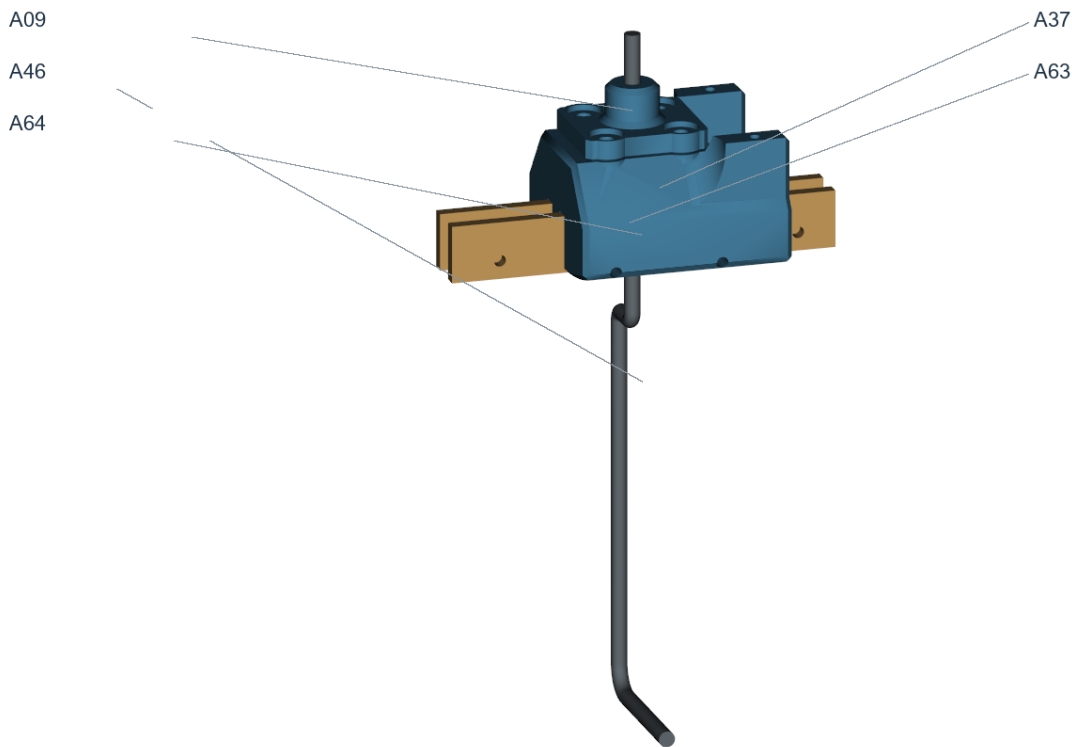
CAD coordinates: X is spanwise, Y increases from nose to tail, Z is up. Nose tip is approximately $Y = -439.65$ mm. The revised assembly STEP is the placement reference. The source typo `Notor_Mount` is the motor mount A26.

1. Lay A62F and A62R on a flat jig along the centreline; first make the nose bypass on page 5. Do not glue them end-to-end across the shaft gap.
2. Fit forward longitudinal member A13 and aft member A43, then former A34 at $Y = -130$, A54 at $Y = -5$, A16 at $Y = 233.65$ and A04 at $Y = 384.65$. Slots determine final seating.
3. Fit the main-gear box A25/A48 with bottom plate A47. Fit tail bulkhead A18 and revised motor mount A26.
4. Check squareness from the assembly axes, equal left/right former extents, and that all tabs seat without forcing. Test fit the printed nose, centre and tail fairings before bonding.

Bond wood joints after the full dry assembly fits. Use modest structural fillets at gear, wing-root and motor load paths; remove excess adhesive rather than adding mass indiscriminately. Keep battery/avionics access open. Stock thickness and laser kerf control the actual fit; resolve a tight joint by the coupon/process setting, not by scaling the DXF.

Revision: overlapping A40/A62 source spines have been consolidated. There is no separate A40 piece to install. The nose shaft requires the intentional A62F/R split and bolted side doublers.

05 / Nose gear bypass



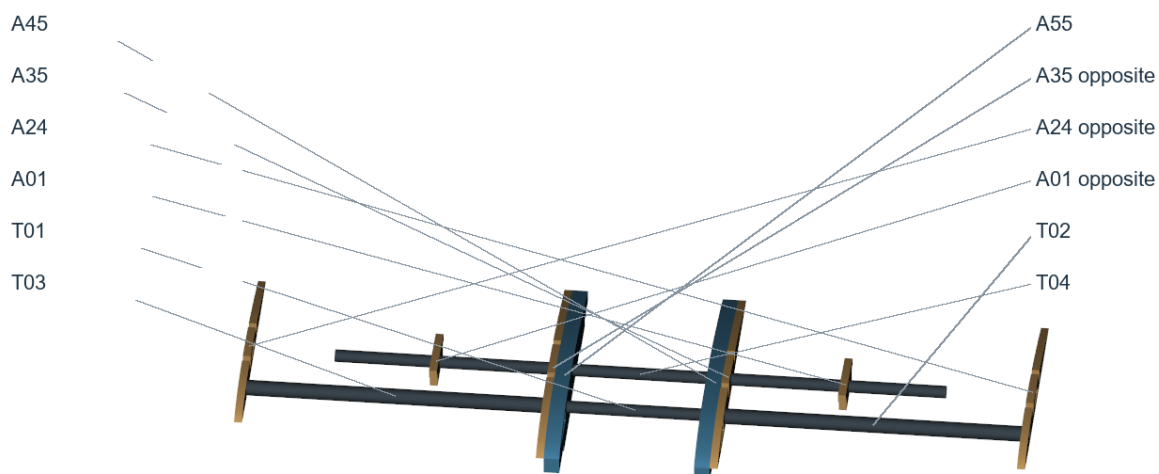
The original 3 mm steering shaft passed through the thin centre spine. The revised joint routes the connection through two continuous 2.5 mm plywood cheek plates A63/A64 and four M2 bolts. A37 has matching recessed pockets. Each cheek is 72 x 10.7 mm; face grain runs along its length.

1. Place A62F/R collinear with the 3.4 mm shaft gap centred on $Y=-293.65$.
2. Fit one cheek on each side of the 5 mm spine. Dry-fit them into the revised A37 pockets.
3. Install four transverse M2 bolts at $Y=-320, -302, -282, -264$ mm, $Z=-37.964$ mm. Inner bolts pass through the mount; outer bolts clamp the wood stack. Use washers and locking nuts.
4. Fit A09 and the extended 3 mm wire. Rotate the shaft fully. Bond wood interfaces only after the assembly stays straight and turns freely; keep glue out of the shaft passage.
5. Tighten enough to seat the stack without crushing plywood/plastic. Recheck after cure.

The corrected wire extends its upper shaft to $Z=+1$ mm, through the bearing and steering-arm region. Fit an actual 3 mm clamp arm/collar and trim only after engagement is established. Do not copy the short, disconnected shaft in the original assembly. The shaft must not rely on glue for retention.

Ground inspection: support the body, steer both directions, move the wheel through its allowed deflection and check the bypass, mount and skin opening together. Inspect for cracks, loose bolt bearing and rubbing before flight. No landing-load rating is asserted by the digital fit check.

06 / Carbon joiners and retention



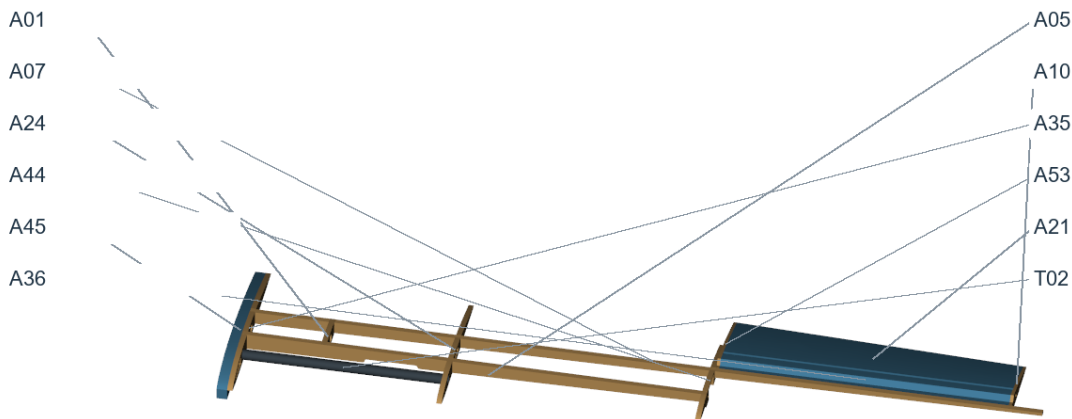
Part	Section / cut length	Position and purpose
T01	8 OD /6 ID; 406 mm	CentreX=-203..203. Front joiner axisY60.454, Z15.632.
T02/T03	10 OD /8.2 nominal ID; 199.35 mm each	RightX53..252.35; leftmirror. Bond sleeves into wing root/middle ribs.
T04	8 OD /6 ID; 390 mm	CentreX=-195..195. Rear axisY100.500, Z15.070.

Buy or have the shop make a matched slip-fit pair. Nominal 8.2 mm sleeve ID gives 0.2 mm diametral clearance around 8 mm joiner; actual tube tolerances govern fit. Do not force unmatched stock or silently sand away structural wall thickness. Plywood sleeve holes are 10.2 mm; root/rear support holes 8.2 mm.

Dry-fit both wings on the centred tubes before bonding sleeves. Front insertion is approximately 150 mm each side into the wing sleeve. Keep adhesive off the sliding joiner. Retain each removable wing using the three aligned M2 root fasteners. Check there is no spanwise escape, twist or hinge interference. Mark insertion positions on tubes.

This replaces overlapping source tube references and recentres the rear tube. Do not cut the source A28 stub or use source A33 as a second 10 mm tube. Your 5 mm solid carbon rods are spare material, not the wing-joiner specification.

07 / Wing frames and ailerons



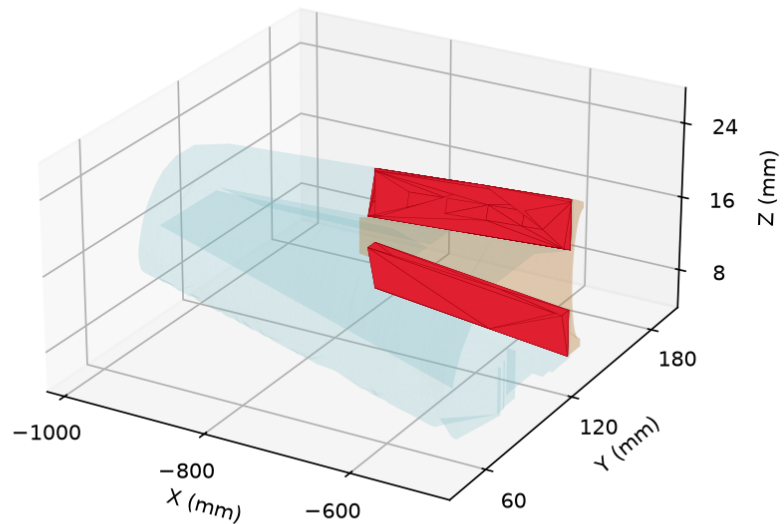
Build one right wing, then the corresponding left-hand mirror. Plywood through-cut profiles are reused at the listed quantity; assembled STEP files show the mirrored positions. Do not build two right wings.

1. Fit inner rib A35, middle rib A24 and outer rib A44 to front spar A05 and rear spar A07. Fit rear joiner support A01 and the front sleeve. Match each rib slot, not a guessed spacing.
2. On a flat jig, check leading/trailing edges and equal incidence before bonding. Finish A24/A44 edges to the marked noncut sanding guide where needed; preserve revised bore centres.
3. Install the aileron end plates A53/A10, hinge rail A36 and aileron A21 on the right; use A23/A11 on the left. Dry-fit smooth 3 mm pins and retain them mechanically, leaving about 1 mm axial freedom.
4. Install each servo with its manufacturer screws and a serviceable mounting pad. Transfer actual servo mounting holes after confirming its size; the CAD servo is a reference envelope.
5. Fit the control horn and neutral pushrod before closing the skin. Check travel with no scraping, stiffness or exposed sharp pin end.

Do not glue ailerons to the fixed rail. Print a moving-fit sample first; finish holes/hinge interfaces to the actual pin with the outside dimensions unchanged. Do not solve a tight hinge by scaling the entire STL.

08 / Foam skins and seams

Wing hinge inner core | red = CAD interference to relieve



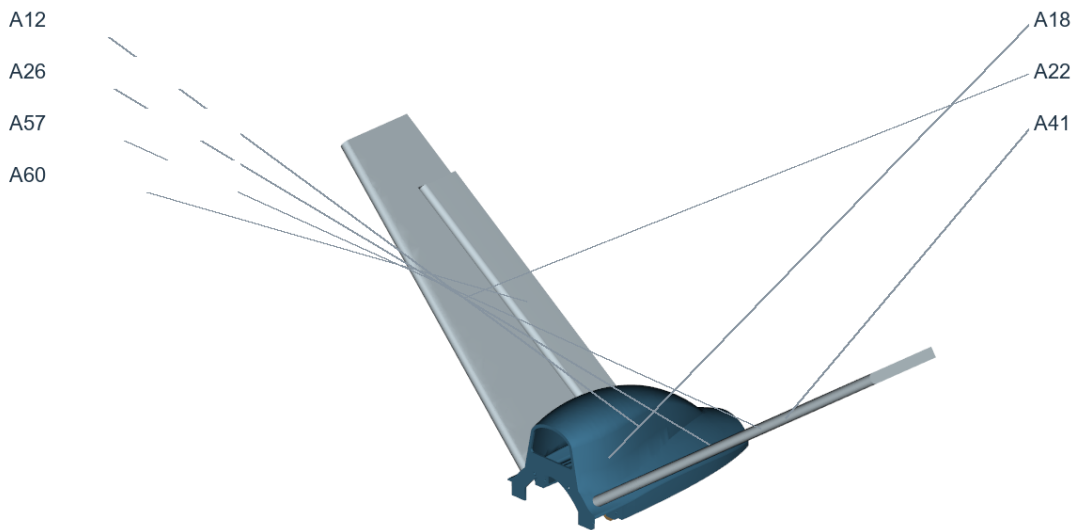
Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

Use the developed F-series patterns and their forming guide. They follow the curved CAD surface. Print at 100%, check scale in both directions, and assemble paper tiles using the common marks/window coordinates and 10 mm overlaps.

1. Mark each outside face and part ID. Knife-cut the finished skin edge. Moving-tail blanks have 5 mm sacrificial shaping stock. F23/F24 include a 1 mm PAPER-ONLY lower seam lip: remove its core and trim/fold it against the mating lip as shown in the foam guide. Other skin CUT lines have zero added lap/trim.
2. Peel inside paper only in the curved forming zones. Keep outside paper intact as the pattern reference. Gradually relieve/compress inside foam and dry-form over the real frame; do not stretch a short paper edge into place.
3. Seat the wing from its leading-edge region toward the open/trailing seam. Shave local inner foam for the printed hinge rail and rib contact patches. Do not cut away the outside paper to hide an inner interference.
4. Bevel inside seam edges to their actual meeting angle. Butt outside paper edges and add 15 mm internal paper straps, about 7.5 mm per side. Reinforce exposed seams with suitable light tape.
5. Dry-fit an entire wing and its moving surface before gluing progressively along the frame. Hold alignment while adhesive cures.

Peeled board loses stiffness and can warp. Use symmetric restraint and prompt bonding to the frame; reinforce seams and relief boundaries without reintroducing an interference. Make the forming/adhesive coupon before cutting the full board set. Use XPS-compatible glue; ordinary CA must not touch exposed XPS unless its manufacturer explicitly approves it.

09 / Fuselage covering and tail



Fit printed nose A59, centre fairing A06 and rear/tail fairing A12 to the dry skeleton. A12 includes the revised bulkhead pocket. Keep a removable battery/electronics access panel rather than permanently closing every cover before the owner has balanced the aircraft.

Match F-series fuselage panels to their source-face diagrams. Install lower forward, lower middle and upper panels in their numbered groups. The rear underside is divided into narrow developed bands: assemble them in parameter/number order and support each butt seam internally. Shave inner-core contact patches against the actual longitudinal wood and spine before bonding.

Cut the main-gear and nose-wire exit only after tracing the installed gear. Start with 0.5 mm around the static leg, then check the required deflection/steering clearance on the ground. The foam must not carry gear loads. Reinforce the paper around the opening.

For each fixed V-tail, use the removable tail-forming collars supplied with the foam tooling PDF to check the exterior at the marked stations; they are tools, not aircraft parts. Form its paired skins/root collar and test slot engagement in A12. The separate control is 10 mm thick: laminate two 5 mm blanks, transfer the finished reference outline and shape the long edges to R5 using the STEP/end reference. Discard the sacrificial 5 mm perimeter allowance.

Make tape hinges that preserve the CAD hinge line and permit free symmetric movement. Fit two tail servos, horns and pushrods with access for adjustment. Do not bridge a hinge with a rigid seam strap or glue. Fit both tails together and compare symmetry to the revised assembly before the joint cures.

10 / Printing and metalwork

Print group	Starting process settings, then qualify on a coupon
A06/A12/A59 shells	LW-PLA preferred for mass. If using available PLA+, start 0.2 mm layers, 2 perimeters and 3-5% infill; record slicer mass and weigh the finished part. Support internal overhangs without blocking sockets.
A11/A21 ailerons; A23/A36 rails	Start 3 perimeters, 10% infill; orient long axis on the bed with supports kept out of hinge bores. Use smooth pin sample and full travel check.
A09/A37/A45/A55 functional parts	Prefer a shop material/process suited to load and temperature; available PLA+ requires inspection/testing. Start 4 perimeters, 30% infill. Orient so tube/bolt loads do not simply split layer interfaces. Do not approve from infill percentage alone.

STLs are millimetres and translated to positive coordinates without scaling. Shop selects final orientation and support strategy. Some parts exceed common 220 mm beds; use a larger printer or have an explicit joint designed. Do not cut a structural STL into arbitrary pieces or scale it down.

Clean support material completely, check every passage and inspect layer bonding. Compare caliper dimensions against the print manifest. Install no cracked, warped or incompletely bonded root bracket. Record total print mass before the owner decides battery placement.

Landing gear

A02 is a formed metal strip reference, 12.7 mm wide by 3.175 mm thick. Its DXF is the formed front view, not a flat laser blank. The metalwork shop chooses an appropriate bendable alloy/temper and real bend radii, preserving the mounting and axle stations. It must calculate its own tooling bend allowance. Do not attempt ideal zero-radius bends from the CAD.

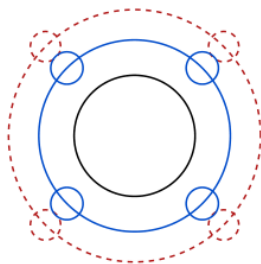
Use the revised two 3.2 mm mounting holes with M3 through-bolts, washers and locking nuts into A47. Fit the three purchased 50.8 mm wheels. Form the 3 mm spring-wire nose leg to the extended STEP reference. These gear references require load/deflection inspection on the completed aircraft; geometry alone does not rate them for a landing.

11 / Motor mount and handoff to electronics

A26 motor mount - declared drilling correction

Source: 4x3mm on 25mm PCD. Revised: 4x3.2mm on 19mm PCD.

Physical Emax motor check required before drilling; drawing below at 2:1.



Red dashed: obsolete 25mm PCD - DO NOT DRILL

Blue: 19mm PCD - four 3.2mm holes

Central 12mm hole unchanged. Minimum radial wood web 1.9mm; use washers.

Revised A26 has four 3.2 mm clearance holes on a 19 mm pitch circle for the Emax ECO II2807. Hole centres are 13.435 mm apart on the square diagonal orientation shown. This replaces the original 25 mm circle. Keep the 12 mm centre opening.

Place the physical motor on the template before cutting/drilling production material. The official drawing is included because marketing wording can be ambiguous. Fit washers and screws with measured safe thread engagement; an overlong screw can reach the windings. Route wires without contact with the rotating bell.

The owned 9-inch propeller is not approved by this fabrication package. The Emax source recommends 6-7-inch propellers, and no verified current/thrust test of the owned 9-inch combination has been supplied. Leave the propeller off during assembly/calibration. The electronics owner must qualify motor/prop/ESC load and ground clearance before flight.

Leave these accessible

Battery tray and straps; FC mounting area; ESC cooling/wiring path; receiver antennas; GPS/compass mount; airspeed tube routing; wing connectors; servo horns; steering joint and gear fasteners. Fit pull strings before skins close. Do not fix the battery position from appearance alone.

Deliver the body with every loose fastener bag labelled, the revised assembly reference, the actual part weights and notes on every fitted/sanded area. The owner uses the separate electronics PDF for wiring, calibration and powered tests.

12 / Checks before the body is signed off

Check	Builder records
Scale and coupons	100 x 20 mm rectangle measured; actual ply/foam thickness; kerf/slot result; printer coupon result.
Structure	All 31 wood pieces accounted for; no source A40 duplicate; bypass bolts installed; no split grain/cracks; aligned centreline.
Wing attachment	Both wings fit simultaneously; matched sleeve fit; centred tubes; six M2 retainers; no spanwise escape.
Controls	Two ailerons, two tail controls and nose steering free; hinge pins retained; horns secured; no glue in moving gaps.
Skins	Full surface dry-fit; seams meet without forced stretching; core relief/gear exits complete; outside paper sound; no wing twist.
Gear and motor	Positive gear fasteners; free wheels; free steering; motor pattern/screw engagement checked; ground clearance examined.
Mass and access	Total body and individual print masses recorded; battery/FC access open; wires can be serviced.
Outstanding evidence	Material forming, adhesive/print strength, gear loading, propulsion and flight checks must be recorded physically.

Balance reference for the owner

The geometry-derived quarter-MAC station is approximately 522 mm aft of the nose tip, or 22.2 mm aft of the front joiner axis. It is a provisional setup mark, not an aerodynamic or flight-approved CG range. Exposed wing area is 0.30445 m² and mean aerodynamic chord 169.83 mm. Mark the datum on the body and leave battery travel available.

Check balance only in complete flight configuration. Moving the battery changes CG by battery mass / total aircraft mass times the movement. Static geometry does not establish elevator authority, stability margin or take-off capability. An experienced fixed-wing pilot must assess the completed configuration before flight.

Digital evidence

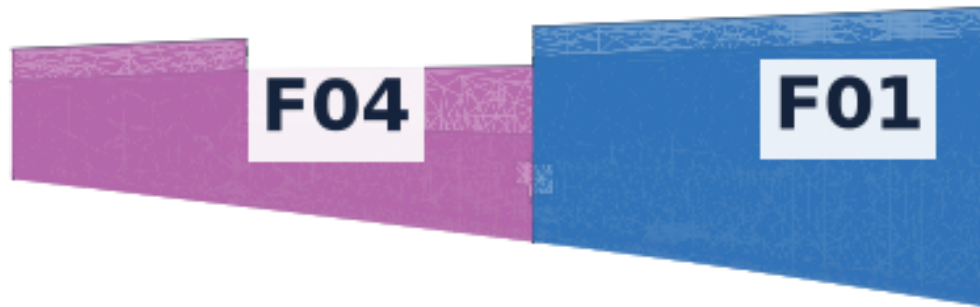
The package includes native profile checks, revised valid solids, printable mesh checks, surface development/seam evidence and numerical PDF scale/registration checks. The Verification folder and README state the scope and remaining physical operations. Original CAD, drawings and transcript remain traceable in Source. All revisions are listed; no global scaling was used.

Sources: supplied Onshape STEP/transcript; official Emax mounting drawing; manufacturer material information. Electronics sources and model-specific wiring references are in the owner guide. Illustrations show actual source/revised CAD; colours identify material groups, not final paint.

Wings: match each skin to its side and span section

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

LEFT AIRCRAFT SIDE (-X)



RIGHT AIRCRAFT SIDE (+X)



 F04 Left outer wing A27

 F01 Left inner wing A52

 F02 Right inner wing A49

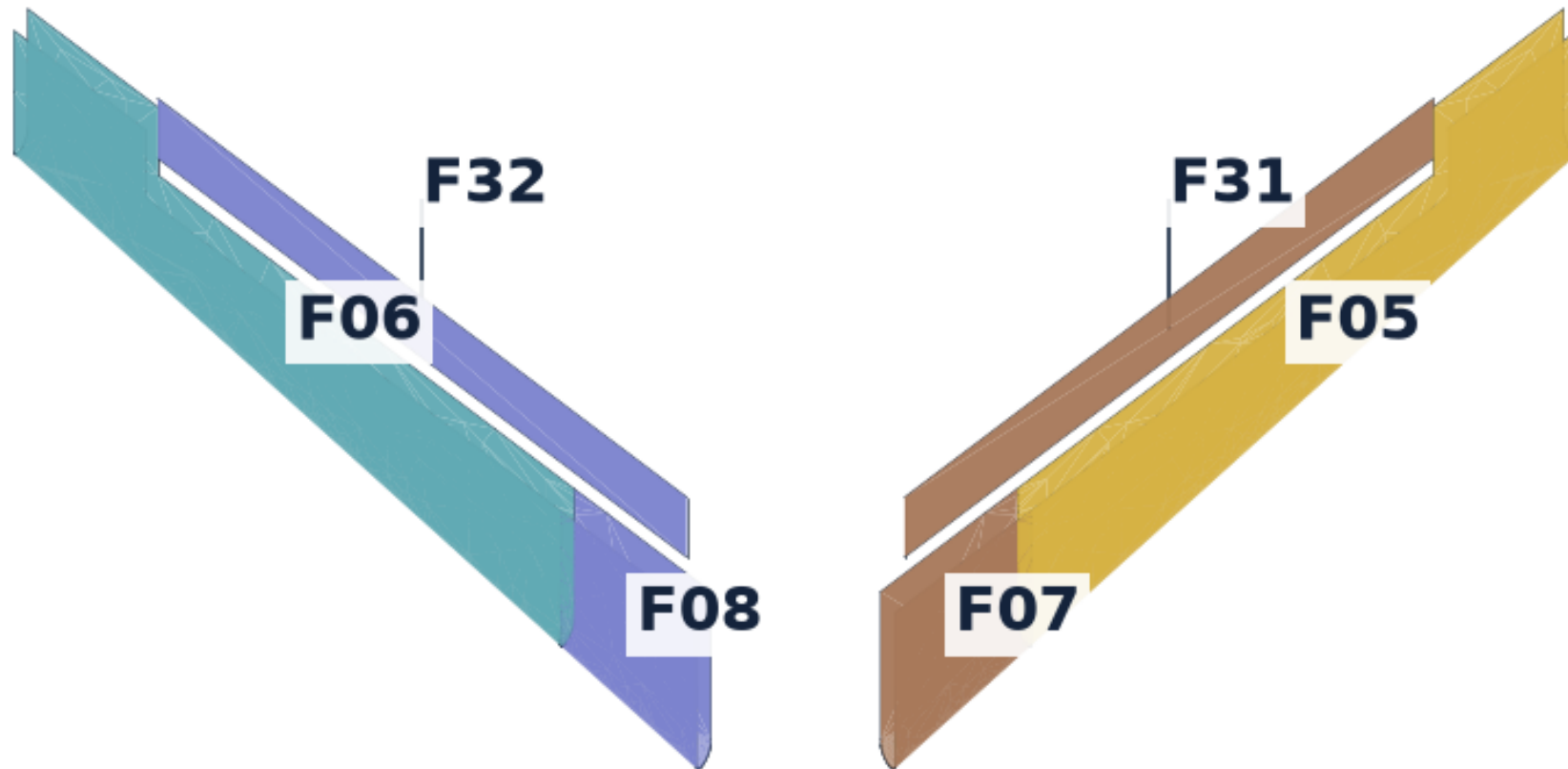
 F03 Right outer wing A38

Fit inner and outer sections to their ribs and spars before closing the trailing seams.

The narrow hinge-rail contact zones need inner-core relief; preserve the outer paper. See the forming guide.

V-tail: two fixed skins and one moving control on each side

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates



- F06 Left main skin A22
- F08 Left root collar A22
- F32 Left control A60: cut two layers

- F05 Right main skin A57
- F07 Right root collar A57
- F31 Right control A41: cut two layers

Each fixed-tail pair meets at the root collar seam (80% of the CAD span parameter from the tip).

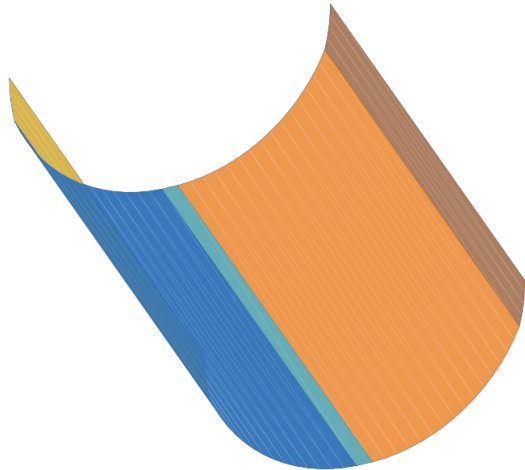
Control highlights show the CAD broad faces. Laminate two 5 mm blanks to 10 mm, then shape the R5 edges.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

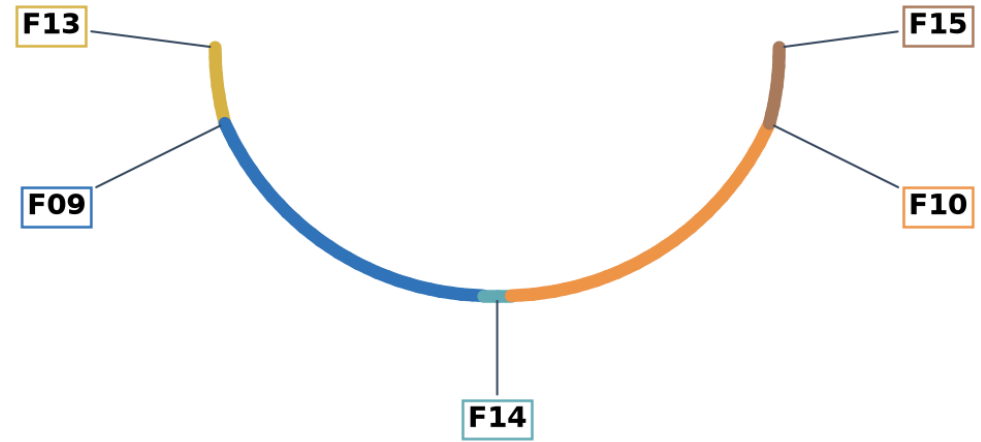
Lower fuselage: five distinct pieces around each section

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

FORWARD LOWER SHELL A56 | Y -335 to -130 mm

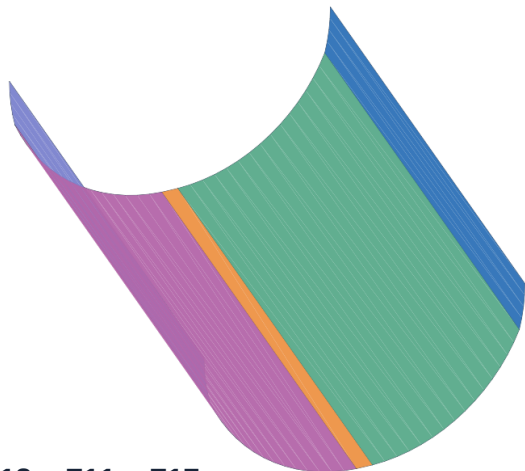


F13 > F09 > F14 > F10 > F15

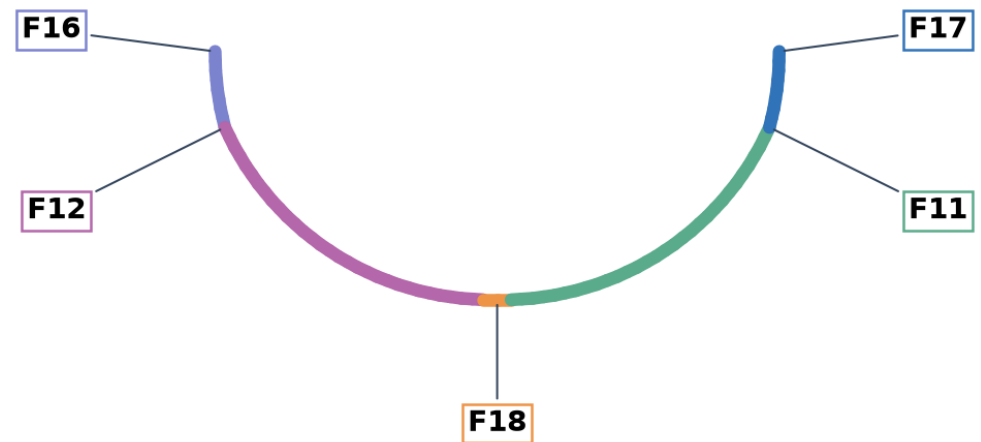


Cross-section: left to right around bottom

MIDDLE LOWER SHELL A17 | Y -130 to +238.65 mm



F16 > F12 > F18 > F11 > F17

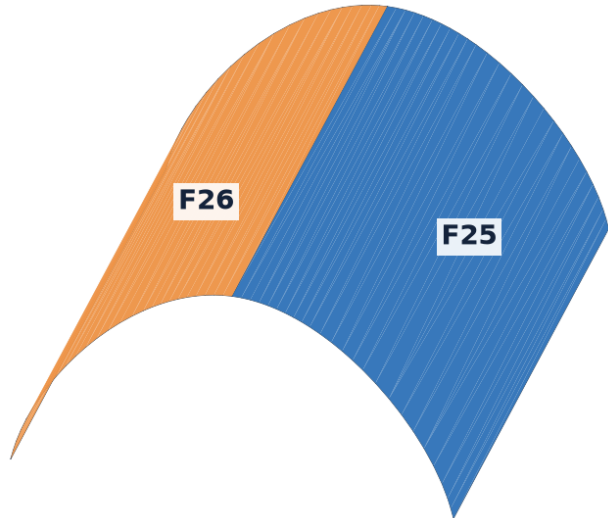


The narrow bottom strips are real parts; retain them.

Upper and aft-lower fuselage: match the coloured face pairs

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

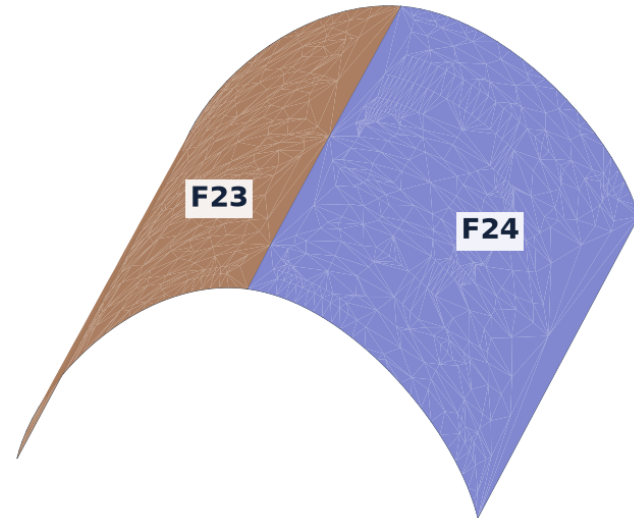
FORWARD UPPER A08 | Y -130 to 0



F26 left / F25 right; meet at the top.

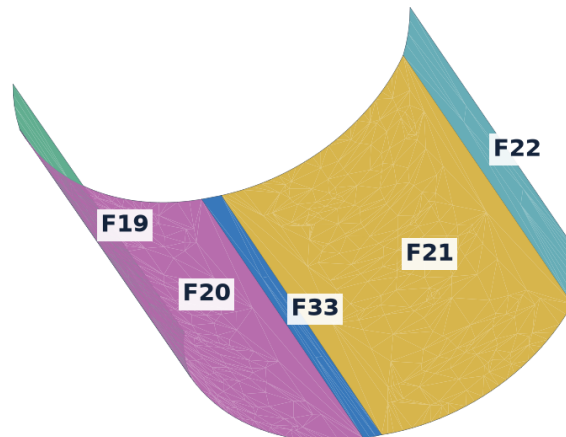
F23/F24 include a 1 mm paper-only lower-edge return: remove its foam, then trim the paper to fit.

AFT UPPER A15 | Y 233.65 to 389.65



F23 left / F24 right; meet at the top.

AFT LOWER A42 | Y 238.65 to 389.65 mm



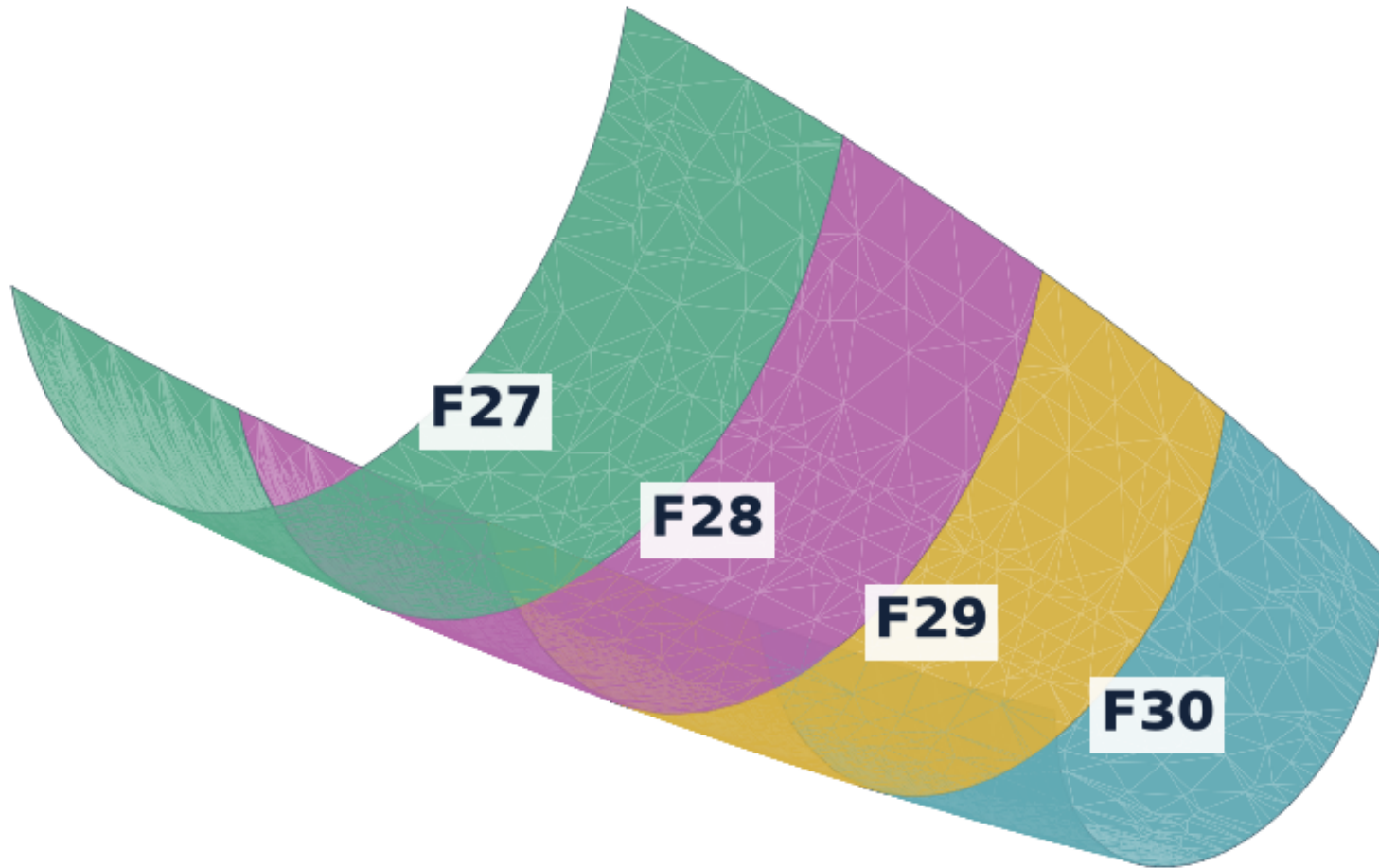
Around the underside: F19 left lip > F20 left curve > F33 bottom > F21 right curve > F22 right lip.

Fit the printed transitions between these shell groups before final skin bonding.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

Rear underside: assemble four bands from front to aft

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates



■ F27 1 - forward band

■ F28 2

■ F29 3

■ F30 4 - aft band

A20 assembly order: F27 > F28 > F29 > F30. The Y coordinate increases toward the tail.

Butt the outside-paper seam lines. Add separate inside straps; dry-fit around A43 and A62 before gluing.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

Foam fabrication method and verification

Design basis

These are quantitatively developed manufacturing surface patterns from the original CAD, in millimetres, without changing scale. They are not top-view wing outlines. Use the confirmed **FliteBoard Pro, 5 mm paper-faced XPS, 400 GSM, 1000 x 600 mm sheets** for the skins. The supplied pack contains ten sheets. The lower forward/middle CAD shell has concentric inner/outer cylindrical faces separated by 5 mm. Tail control surfaces have two parallel broad faces 10 mm apart and 5 mm radius edge surfaces; make each from two laminated 5 mm layers. 400 GSM is an areal mass specification, not a thickness. Measure actual stock thickness and check its batch tolerance against 5 mm before fitting. Do not resize the patterns to accommodate another thickness.

The manufacturing reference is the **preserved outside paper**. On curved skins remove the inside paper, progressively relieve/compress the inner foam, and keep the outside paper intact. Removing inner paper reduces panel stiffness and can cause warp: dry-form and bond to the ribs/formers promptly, restrain the panel symmetrically while adhesive cures, and add compatible paper/tape reinforcement across internal seams. Do not assume this reproduces the stiffness of intact two-paper board. This brings the bending reference close to the CAD exterior. It is not a conventional mid-thickness neutral-axis bend calculation. Leaving both paper skins intact and forcing a tight curve is a different process and will not preserve these dimensions. A paper-faced foam material that cannot form this way requires a new process qualification, not scaling.

Cutting, seams and allowances

DXF units are mm. CUT is a full-depth knife outline. REFERENCE and LABEL are pen marks, not laser cuts. No kerf compensation is embedded: a knife follows the line; a shop using another qualified process applies its measured tool compensation. Default to knife cutting this XPS board. Use a laser only if the institution explicitly approves this exact XPS product and its extraction/process. Avoid CA adhesive unless the adhesive manufacturer explicitly confirms XPS compatibility; qualify a compatible adhesive on scrap.

Skin CUT lines are finished outer-paper seam lines: **zero lap allowance and zero trim allowance**, except the explicitly supplied 1 mm paper-only lower-edge trim lips on F23/F24. The REFERENCE outline on those two parts is the nominal exterior face: remove foam from the added lip, retain the paper and trim/form it to close the tiny shoulder return against the adjoining lower-shell lip. This is not an extra 5 mm foam strip. Butt adjacent outer-paper edges; relieve only the inside foam to clear its neighbour, and tape/glue the inside. Add separate 15 mm wide scrap-paper seam straps where accessible, nominally 7.5 mm each side of the join; these straps are not additional dimensions on the outside skin. Do not double the skin at seams. At a curved junction, shave the inner foam gradually to the actual dry-fit angle while preserving the finished outside edge. A universal 45-degree bevel is not appropriate for these changing tangent angles. Knife and glue gaps are established on the forming coupon.

Tail control CUT outlines include **5 mm sacrificial shaping stock all around** the nominal broad-face REFERENCE outline. Cut quantity two for each control, laminate to 10 mm total, transfer the reference outline and fit the original end planes, then shape the cylindrical long edges to radius 5 mm. The 5 mm all-around blank extension is sacrificial stock, not extra flight-control span or chord. Final control shape must follow the supplied original control CAD. Do not glue a control to a fixed stabilizer or bridge the hinge clearance with a skin strap.

The rear underside is divided into four bands along the CAD surface parameter v . These are deliberate relief seams, not stock-size scaling. The manifest records each panel source and zero-based CAD face. Original CAD surface boundaries provide the other panel seams. Corresponding source XYZ points identify mating surfaces; use the Reconstruction CSVs and the numbered corner marks, not PDF page edges, to locate parts. Letter A/B identifies a source occurrence, not an assumed left/right orientation: match the source STEP to the assembly mapping.

Forming and assembly sequence

1. Verify stock thickness with callipers and make a 100 mm printed scale check in both directions. Print PDF at Actual Size / 100%, disable fit/shrink. The index page has a 100 x 20 mm calibration rectangle; check both directions before printing the complete set. The optimized PDF uses 5 mm hard margins and groups multiple IDs into six posters. Tape only pages with the same S-number, using their row/column labels, matching crosses and 10

mm overlap; do not align by printer paper edges. The index identifies every pattern that is whole on one page. Use the F31/F32 paper templates twice for their two laminate layers. 2. Cut a scrap coupon with a preserved outside paper skin, peel its inside paper and practice forming over the tightest intended nose/leading-edge curve. Gradually remove or compress inner foam. The outside paper must remain continuous, uncracked and close to the CAD surface. Establish glue and seam behaviour here before cutting the final board. 3. Build and align the corresponding CAD ribs/formers and spars first. Check the original assembly's span, root/tip chord and incidence. Skins are surface coverings; the unfolding calculation does not establish structural adequacy. 4. Trace and cut each numbered panel. Transfer its registration marks. Mark OUTSIDE on the paper before peeling inside paper. Keep mirrored source occurrences separate. 5. Dry-form wing skins over their matching rib set from the leading-edge region toward the open/trailing seam. Press progressively along the span, avoiding local paper tension. Relieve inner foam where it interferes with ribs or tight curvature. Do not pull a short seam into place by stretching the outside paper. 6. Dry-form each fixed tail skin using the three removable split collars in Tail_Forming_Profiles.pdf and TOOLING_DXF. They are derived from the actual trimmed CAD solid; use T03 at 6.84 mm from the root, T02 at 171 mm, and T01 at 335.16 mm along the 342 mm span axis. Use stiff card or 3 mm scrap plywood, align their planes on a straight jig board, then remove both halves after the skin seam adhesive cures. No internal fixed-tail ribs were present in the source, so these are temporary external forming tools, not existing aircraft structure. Respect the separate moving control's hinge line and required clearance; fit and test control travel before closing. 7. Fit lower forward and middle fuselage pieces on the matching formers. The narrow side and centre strips are explicit parts, not extra trimming waste. Match the broad cylindrical pieces to those strips using their source face boundaries. 8. Fit the aft lower and upper panels. Install the four rear-underside bands in parameter order 1 through 4, dry-matching both boundaries of each band before gluing. Support each seam internally. Small boundary differences from the numerical approximation must be distributed through dry forming; do not accumulate them at one end. 9. Laminate and shape the two tail controls as described above. Dry-fit hinges and full travel, then fit servo linkages. Protect hinge gaps from glue and covering. 10. Close skins only after checking access, wires, spar fit, control motion and mass/balance provisions. Weigh finished panels and the complete airframe. The digital checks do not constitute a flight release.

Numerical method and evidence

The exterior STEP faces are tessellated with a explicit absolute 0.2 mm deflection / 0.15 rad angular tolerance (BRepMesh_IncrementalMesh with relative meshing disabled). Cylinders use the exact development $x = \text{radius} * u$, $y = v$. Their reported comparison to 3D mesh chords includes arc-versus-chord difference; the continuous cylinder itself has zero intrinsic development strain. Other surfaces use nonlinear least squares of intrinsic mesh edge lengths, with the actual absolute-mesh edge lengths as constraints. The final pattern preserves the CAD surface approximately; it is not a claim that compound-curved foam can become exactly flat without strain.

Compound rear bands are clipped from the same parent surface mesh, preserving shared 3D seam vertices. The output lists every flat vertex, corresponding source XYZ/UV and triangle connectivity. Edge residuals are recalculated from those actual exported coordinates. Final checks include a simple non-self-crossing boundary, no reversed triangles, each panel fitting a 1000 x 600 mm stock rectangle, and less than 0.5% maximum mesh-edge length change (final maximum 0.3665%). This is a numerical design tolerance, not a measured foam strain limit. The CSV/JSON manifest includes 99th-percentile edge error and maximum absolute edge error. Surface interpolation and physical forming introduce additional error; inspect the dry fit rather than relying solely on the percentage.

Scope and remaining physical checks

Original CAD is preserved. These files cover exterior wing and fixed-tail skins, the identified fuselage shell surfaces, and the two 10 mm laminated tail-control blanks. Printed transitions, nose, hinges, wood structure, spars and hardware are covered by the main fabrication package rather than these skin files. No aerodynamic, structural, adhesion or flight performance has been established by unfolding. A shop can cut the patterns at scale and make the specified forming coupon; the builder must inspect actual material behaviour and dry fit before committing the full airframe.

Tail skins are each divided at normalized CAD span parameter $v=0.8$. Panel rootcollar1 covers 0-0.8; rootcollar2 covers 0.8-1.0 at the root. This separates the root transition into a supported collar seam and avoids imposing its full curvature on one sheet. Assemble each matched pair on its CAD stabilizer shape before final trimming of

interior foam.

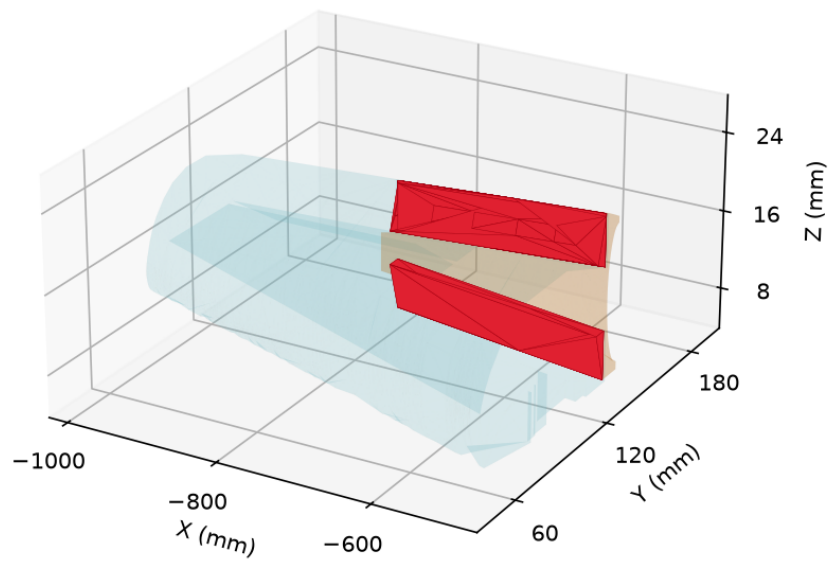
Required local core relief found in assembly

These CAD interferences are explicitly accounted for during dry fitting, before bonding skins. Use the supplied Relief_Reference STEP overlap volumes and illustrated views to locate each zone. They are nominal CAD overlap references; actual removal is fitted against the real component. Do not subtract the entire component bounding box from foam.

- **Outer wing A27 / hinge rail A23**, and the opposite **A38 / A36**: locally relieve the inner foam under the hinge rail. For A23 the assembly footprint is approximately X -770.294 to -495.881, Y 127.001 to 143.096, Z 6.119 to 22.767 mm (actual A27/A23 intersection bounds). The reference overlap volume locates the interference, not a full-depth rectangular hole. Lay the actual hinge rail in place, transfer its contact footprint to the inner foam, and shave only contact regions until the rail seats and its moving hinge runs freely. Keep the outside paper intact. Check the mirrored opposite rail by the same method; do not assume separated-source coordinates are assembled coordinates. - **Lower centre skin A17 / landing gear A02**: the gear exit needs a local through-opening. Trace the actual fixed leg where it crosses the skin, cut only that exit, and provide an initial 0.5 mm cutting clearance on each side. This is an assembly clearance, not a demonstrated landing-deflection clearance. Check full gear deflection on the ground with the aircraft supported and enlarge only as required so the skin does not carry gear load. Reinforce the paper around the exit with compatible paper/tape bonded to sound surrounding skin; preserve access to the actual structural gear attachment. - **Lower centre skin A17 / root brackets A45 and A55**: mark each actual bracket contact patch on the inside and relieve only the inner foam. Preserve the exterior paper. Do not sand down structural brackets or wood to make the skin fit. - **Rear underside A20 / longitudinal wood A43 and spine A62**: dry-assemble both wood members before fitting rear bands F27-F30. Transfer contact patches into the inner foam and relieve progressively. Preserve the outside paper and avoid routing a relief across the rear-band paper seam without a separate interior paper/tape bridge.

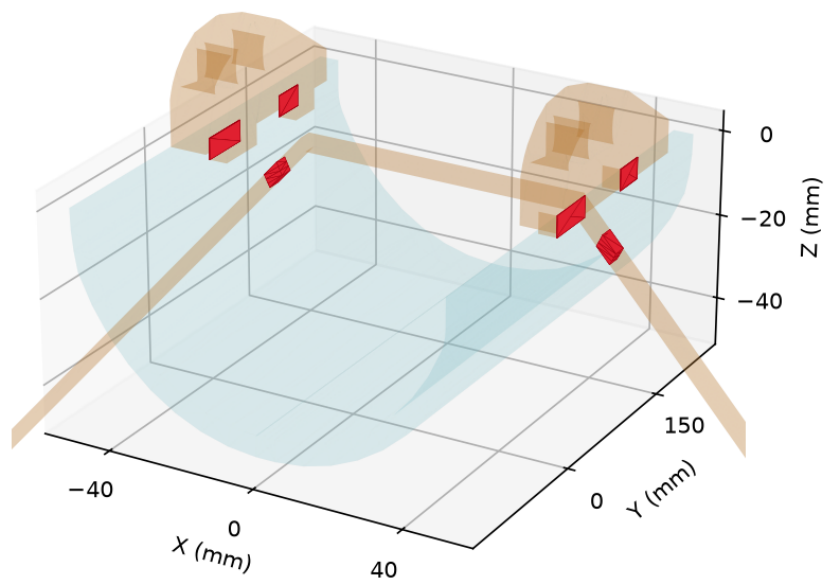
After these operations, repeat the dry assembly with all affected parts installed simultaneously. A skin can fit each neighbour independently while still binding when the complete stack is assembled. Re-laminate compatible paper/tape at inner seams and relieved-zone borders after the geometry fits; avoid adding material that recreates the collision. The stiffness of peeled/relieved foam has not been established by the CAD check.

Wing hinge inner core | red = CAD interference to relieve



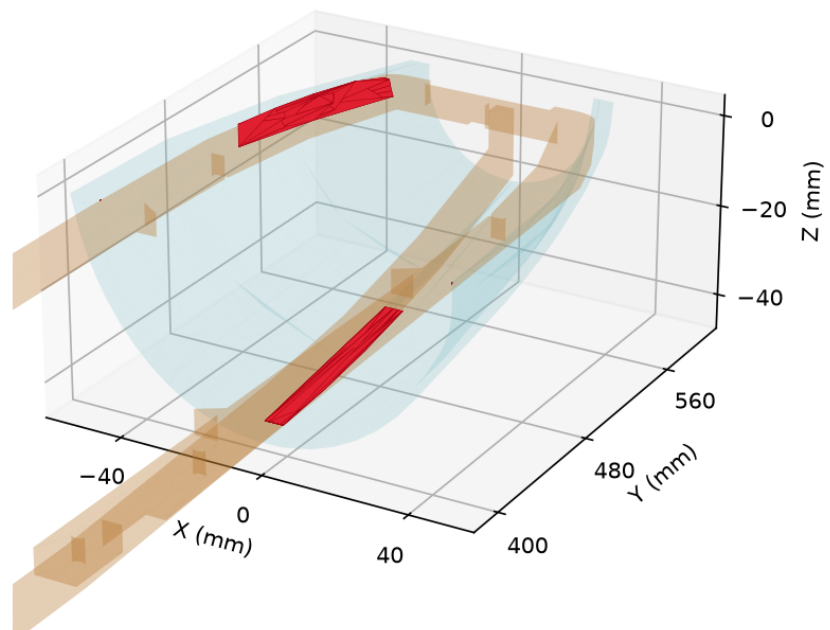
Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

Lower center inner core | red = CAD interference to relieve



Blue: A17 foam. Tan: A02, A45, A55. Red: trim contact region. View cropped; axes not equal scale.

Rear underside inner core | red = CAD interference to relieve



Blue: A20 foam. Tan: A43, A62. Red: trim contact region. View cropped; axes not equal scale.

Final absolute-mesh audit

The earlier relative-meshing trial was rejected and is not included. Every released pattern has been regenerated from an explicit absolute 0.2 mm surface mesh. The `Surface_Mesh_Validation.json` report compares mesh area with adaptive CAD surface integration and samples the centroid and all three edge midpoints of each triangle against the exact STEP surface. Maximum sampled normal departure is 0.19971 mm. This is a sampled geometric check, not a full Hausdorff-distance proof. Maximum area difference is 0.315% for the complex rear shell; other curved parents are within 0.068%. Added tail collar seams have matching developed lengths; the largest rear-band seam mismatch is 0.02667 mm.

Assembly map and exterior-face coverage

Use `Foam_Assembly_Map.pdf` and `Foam_Assembly_Mapping.csv` to locate all 33 pattern IDs. The aft-lower shell A42 has five pieces around its underside: F19 > F20 > F33 > F21 > F22. F33 is the real narrow bottom strip; do not omit it. `Source_Face_Coverage.csv` records the route for every face of the foam source components. Interior faces and material-thickness edges are not additional skin panels. A15 sub-millimetre outer shoulder returns are covered by the paper-only trim lips on F23/F24.

The fixed-tail forming templates confirm there is no useful cavity after a nominal 5 mm inward offset. Shape and relieve the two foam walls so they meet inside while retaining the outside paper; do not add unmodelled internal ribs. The external split collars are removed after cure.

Efficient paper printing

The foam template PDF now contains 22 cutting pages plus one index (23 total), replacing the earlier 72-page layout. All 33 source DXFs are unchanged. Every pattern that fits a single usable A4 page is complete on at least one indexed page. Small pieces occupy clear spaces outside the large wing/tail cut polygons. The tail-tooling PDF uses one instruction page and one shared cutting page (2 total); all six removable collar halves are whole on the cutting page. `Foam_Paper_Layout.csv` and the fit/registration verification files record placements and

source-to-PDF checks.

Electronics build guide

Matek F405-WING V2 / ArduPlane / FlySky / five servos with nose steering

A bench-to-airframe sequence for the supplied MQ9-style model. It defines a wiring plan and acceptance checks. No aircraft was connected, calibrated or flown while preparing this guide.

Remove the propeller for wiring, firmware, calibration, servo and failsafe work. Refit it only for a separate controlled propulsion test after bench checks pass.

Equipment	Build assumption
Flight controller	Matek F405-WING V2; not V1, WTE or TE
Navigation	Matek M9 GPS; exact SKU and compass presence unconfirmed
Controls	Two aileron + two V-tail + one nose-steering servo for supplied gear; 9 g metal gear, model unknown
Power	GenX 4S 5200 mAh LiPo; ReadytoSky 40 A 2-4S ESC advertised as Opto with 5 V / 3 A BEC
Propulsion	Emax ECO II 2807 1300 KV + Gemfan 9045 CL three-blade; combination unqualified
Radio / airspeed	FlySky FS-i6X + FS-IA10B; Matek airspeed SKU unconfirmed

Work in this order

Identify parts -> build and meter harnesses -> configure firmware -> connect radio and sensors -> centre and test servos -> test failsafes -> install and balance -> qualify propulsion. Save a parameter file after each successful stage.

This guide does not assign a flight CG, control throw, stall speed, return altitude or flight gain. Those require the completed airframe, aerodynamic review and measured tests. The supplied CAD includes nose steering (A39); fit five servos for that build. Flaps and payload servos are outside this plan.

Resolve these labels first

Connector shape and wire colour are not a pinout

Unknown	Safe branch before connection
M9 GPS SKU	Read PCB and module labels and obtain that Matek manual. UART module: TX to FC RX3, RX to TX3; match specified supply and ground. Add SDA/SCL only if an I2C compass exists. CAN-only module: stop; this FC has no native CAN. Never infer pin order from another M9.
Airspeed SKU	ASPD-4525: I2C, ARSPD_TYPE=1. ASPD-DLVR: labelled I2C connector, ARSPD_TYPE=9; its CAN connector is different. Verified I2C branch: DA2/SDA + CL2/SCL, ARSPD_BUS=1. Other SKU: use its manual and leave disconnected until identified. [1,6,7]
ESC signal plug	Determine whether red is BEC output, absent, or required optocoupler supply. If BEC output, depin and insulate individually; retain signal and signal ground. If input-side power is required, follow the exact manual. Do not guess from Opto in a listing.
9 g servo model	Confirm voltage, pulse range, current and torque. Use 5 V only if supported. Do not select 6/7.2 V from size or metal gears alone. Five servos cover the supplied controls and steering; quantity is not torque approval.
Battery / connectors	Record mass, dimensions, connector polarity and charge limits. Meter every adapter. Never connect the 4S main lead to receiver, GPS, servo rail or 5 V pad.

Make a connector record

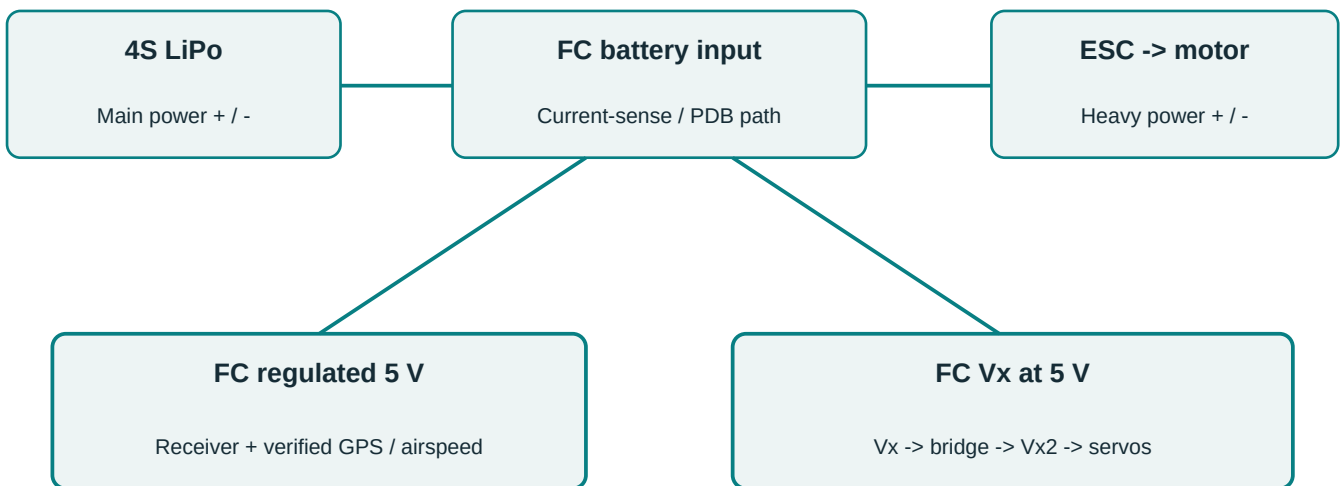
Sketch each plug viewed from a stated face; number cavities; write signal names; continuity-check both ends with all power removed. Label the completed harness. A keyed plug can still be wired incorrectly.

The remaining pages name functions and board pads. They intentionally do not assign cavity numbers for unidentified hardware. Complete this identification before soldering the peripheral harness.

Power architecture

One positive source per rail; common signal ground

Route propulsion current through the FC power-distribution path. This is a functional diagram, not a connector drawing. Match battery-input and ESC-output pads against the V2 layout [1].



ESC signal + ground -> FC S1 + G

ESC BEC output (if present) -> insulated; never joined to FC rails

Build and meter

1. Fit strain-relieved battery and ESC power leads to the documented pads. Keep high-current loops short. Select connectors and wire for measured load; insulate exposed joints.
2. Keep electronics 5 V and servo Vx positives separate. S5-S9 use Vx2: bridge Vx to Vx2 for this internal-BEC plan. Check before plugging in servos. [1]
3. If tests require an external BEC: remove the bridge, verify isolation, power Vx2 from that BEC and move all servo positives to that rail. Retain grounds; never join two BEC positives.
4. With peripherals unplugged, check polarity and hard shorts. First power through a suitable current limiter, then meter 5 V and Vx/Vx2 at far connectors. A limiter may brown out under servo load; it does not qualify flight power.

Wire the signal harness

FC outputs and transmitter channels are independent

FC connection	Destination / setting
S1 + G	ESC PWM signal + ground. SERVO1_FUNCTION=70 (Throttle). Isolate BEC output lead if present.
S2	Unused, SERVO2_FUNCTION=0; leave empty.
S3 + Vx + G	Left aileron. SERVO3_FUNCTION=4 (Aileron).
S4 + Vx + G	Right aileron. SERVO4_FUNCTION=4 (Aileron).
S5 + Vx2 + G	Left tail. Initially SERVO5_FUNCTION=79 (V-tail left). Validate page 8; physical motion governs.
S6 + Vx2 + G	Right tail. Initially SERVO6_FUNCTION=80 (V-tail right). Validate page 8.
S7 + Vx2 + G	Nose steering A39: SERVO7_FUNCTION=26 (GroundSteering). Omit only with a deliberately fixed nosewheel.
S8-S9 / LED	Unused; disconnect. Check SERVO8-10_FUNCTION for unwanted assignments.
RX2 + 5 V + G	FS-iA10B iBUS servo-output signal + supply + ground. Not SENS telemetry port. BRD_ALT_CONFIG=0.
RX3 / TX3 / supply / G	Verified UART GPS: module TX -> RX3, RX -> TX3; supply per exact SKU. SERIAL3_PROTOCOL=5 (GPS).
DA2 / CL2 / supply / G	Verified I2C compass / airspeed: SDA -> DA2, SCL -> CL2. Match pinout and supply. Check shared-bus address conflicts.

This custom allocation replaces standard Plane outputs 1/2/3/4 (aileron/elevator/throttle/rudder). An S1 label alone does not mean throttle. Write and read back each assignment before attaching the ESC. The hardware definition maps S1-S9 to PWM1-PWM9. [2,3,4]

Use conventional PWM initially at 50 Hz (SERVO_RATE), only if supported by identified servos and ESC. Keep DShot disabled for this unknown ESC. Shared timer groups constrain rate/protocol changes. [2]

Firmware and first connection

Mission Planner on Windows; menu names can change

Install the matching target

1. Photograph the board and confirm F405-WING V2. Obtain Mission Planner from ArduPilot. Use a data-capable USB cable with battery, ESC and servos disconnected.
2. Choose stable **ArduPlane**, target **MatekF405-Wing**, version 4.4 or newer. Do not select Copter, INAV, F405-STD or F405-TE. V2 requires newer sensor support. [1]
3. With an ArduPilot bootloader, use SETUP > Install Firmware or Load custom firmware for the matching .apj. Without one, follow the board first-install/DFU procedure using matching firmware containing a bootloader. A missing serial port does not prove the board is dead. [2]
4. Connect; record firmware version, target and vehicle type. Save initial full parameters. Review sensor/storage messages. Insert a supported microSD for logs with power off.

Configure before powering outputs

Use CONFIG > Full Parameter Tree/List for page 4 settings. Write, reboot, reconnect and read back. Use SETUP > Mandatory Hardware for accelerometer, compass, radio and modes; Servo Output for output settings. Parameter names govern if menu wording differs.

Do not bulk-import an internet parameter file. Keep arming checks enabled. Resolve each failed check instead of setting ARMING_CHECK=0. Save the parameters actually read from this board.

Recovery route

Disconnect propulsion power, reconnect USB and compare with the last saved parameter file. Revert the changed settings. After firmware changes, repeat output, sensor and failsafe verification. For a non-booting board, use its documented bootloader recovery with the exact target.

Nose steering: centre A39 with SERVO7_TRIM, limit endpoints mechanically, then set reversal so MANUAL right rudder steers right. Check by pushing the model with motor off. GroundSteering=26 recentres above GROUND_STEER_ALT. For stabilized ground steering, establish a healthy compass, set the altitude gate deliberately, and measure turning diameter for STEER2SRV_P; tune using the official guide before powered taxiing. [14]

Radio input and modes

Plain controls from FS-i6X; mixing inside ArduPlane

Create a new airplane model, select AFHDS 2A and bind the FS-iA10B per its instructions. Disable transmitter V-tail/elevon and custom mixes. Centre trims. Confirm iBUS on the receiver servo serial port. FS-iA10B accepts regulated 5 V; its specification allows 3.5-9 V. [5]

Input	Intended function
CH1	Roll; RCMAP_ROLL=1
CH2	Pitch; RCMAP_PITCH=2
CH3	Throttle; RCMAP_THROTTLE=3
CH4	Yaw; RCMAP_YAW=4
CH5	Three-position mode switch; FLTMODE_CH=5. Measure its actual detent values.
CH6 optional	Dedicated arm/disarm only after verifying the installed firmware option and behaviour; never share with mode control.

Calibrate

Power RX from verified FC 5 V; USB alone may not power it. In Radio Calibration, check that each stick moves only its intended bar. Move sticks and assigned switches through full range; finish throttle low, other sticks centred. Save. [8]

Check RCn_REVERSED semantics: increasing normalized input means roll right, pitch up, more throttle and yaw right. Correct inputs first; physical surface direction is adjusted later using SERVOx_REVERSED. Recalibrate after transmitter endpoint changes. [8]

In Flight Modes, set all six slots MANUAL initially; assign the slot reached by the middle detent to FBWA. Keep the third detent MANUAL until navigation is validated; deliberately assign RTL later. Verify HUD mode at all detents and transition bands.

Receiver voltage telemetry is not automatically propulsion-pack voltage. Separate receiver antennas from carbon, battery and motor wiring, and position active ends per the radio manual. Perform the documented range test after final installation.

Sensors and battery readings

Calibrate in final orientation; inspect the actual response

Attitude, heading and position

Mount FC securely and set AHRS_ORIENTATION to match. Run six-position accelerometer calibration, then level calibration on an accurately level aircraft datum. Roll and pitch the aircraft; verify HUD attitude follows correctly before stabilized-mode tests.

Determine whether GPS includes a compass; FC has none onboard. Set external compass orientation and calibrate away from steel, magnets and power wires. Point the aircraft in several known directions and compare heading independently. Repeat after final installation. [2]

For verified UART M9, use SERIAL3_PROTOCOL=5 and an appropriate GPS type (usually Auto for a u-blox UART module). Acquire outdoors with clear sky. Verify plausible position, fix and time. A power LED is not navigation health; GPS-only hardware does not provide magnetic heading.

Battery monitor - V2 values

Parameter	Initial value
BATT_MONITOR / voltage / current pins	4 / BATT_VOLT_PIN=10 / BATT_CURR_PIN=11
BATT_VOLT_MULT / BATT_AMP_PERVLT	11.0 / 66.7; older-board 31.7 is wrong for V2
BATT_CAPACITY	5200 mAh nominal; validate usable capacity separately

Compare reported voltage with a reliable meter simultaneously. Correction: new multiplier = old multiplier x meter voltage / reported voltage. Compare current with a calibrated inline instrument at steady meaningful load; new current scale = old scale x reference current / reported current. Do not fit scale near zero. Check idle offset and consumed capacity through a measured discharge. [1,4]

Choose low/critical thresholds, reserve, delays and actions from pack specification and loaded behaviour, then record on page 11. C-rating does not prove motor current, usable capacity or endurance.

Servos and V-tail direction

Left/right viewed from behind the aircraft, facing forward

Centre each servo electrically before fitting its horn. Align linkages mechanically, then use SERVOx_TRIM for small corrections. Establish conservative endpoints and inspect every hinge and pushrod through travel. No generic travel angle is assigned.

Test	Required response
MANUAL: roll right	Right aileron trailing edge up, left down; reverse for roll left.
MANUAL: pitch up	Both tail trailing edges move in upward pitch-producing sense; reverse for pitch down.
MANUAL: yaw right	Both tail trailing edges move toward aircraft right; reverse for yaw left.
FBWA, centred sticks: tilt aircraft right	Left aileron up, right down: correction opposes tilt.
FBWA: raise nose	Tail command produces nose-down correction; lowering nose produces nose-up correction.

CAD inspection shows an upright V-tail. Initial 79/80 assignments still require physical direction checks. Per channel: wrong in both pitch and yaw -> reverse output; correct for one axis only -> exchange function 79/80 and repeat. Final function labels may not match physical side. [9]

Use the V-tail roll-to-rudder check only with nonzero KFF_RDDRMIX and restore its intended value afterward. A stationary yaw-rotation test is not a heading-hold test. [9]

Combined commands and supply loading

Apply simultaneous full pitch/yaw in all four combinations. Check binding, over-centre horns, pushrod buckling, buzz and saturation. Adjust mechanics, endpoints and MIXING_GAIN together. Repeat MANUAL and FBWA tests after each change. [9]

Exercise all installed servos together under representative load without prolonged stall. Capture Vx and 5 V minima, current peaks, temperature and resets. A multimeter can miss brief dips. Vx is rated 5 A continuous / 6 A peak; electronics 5 V is 2 A. These are ceilings, not measured margin. [1]

Airspeed and propulsion

Sensor response is not calibration; motor spin is not thrust

Airspeed branch

After SKU verification, connect page 2 I2C harness, set ARSPD_TYPE and ARSPD_BUS, reboot and check health. ASPD-4525 uses type 1; DLVR-L10D I2C uses type 9. Never connect the DLVR CAN socket to I2C. [6,7]

Put pitot in clean airflow away from prop wash. Connect dynamic/static lines per probe manual, without kinks or leaks. Allow ambient-temperature settling. Loosely shield pitot from wind without sealing ports; run Preflight Calibration for zero. [10]

Use gentle airflow toward probe and verify a positive response returning near zero. Avoid strong pressure, liquid or blowing into tubing. This proves response only. Keep ARSPD_USE=0 during initial validation, but some current Plane versions still use a healthy enabled sensor for speed scaling; this is not complete isolation. [10,11]

If suspect, disable sensor (ARSPD_TYPE=0) and resolve before flight planning. Ratio calibration requires an independently safe aircraft and controlled flight/log review. A bench airflow check cannot establish true airspeed or stall speed.

ESC and motor

Mounting: the Emax drawing calls out four M3 holes on a 19 mm pitch circle (13.435 mm adjacent square spacing), despite marketing saying 19x19. Check the actual motor with a paper template before drilling. Select screw length from mount thickness and measured thread depth; screws must not reach windings. Emax lists 6-7 inch props; the supplied 9-inch combination needs measured qualification. [15]

Confirm ESC manual and PWM range. High/low calibration requires FC and receiver to remain powered while ESC power is separately switched. USB may not power this RX rail: use a documented isolated setup or separate servo tester; do not backfeed a rail. [12]

Only if confirmed by ESC manual: prop removed, ESC unpowered, command maximum; power ESC, await specified calibration indication, command minimum, verify acceptance. Otherwise follow its documented method. Mission Planner multicopter motor calibration is not for Plane. [12]

Verify motor stopped when disarmed and at minimum. Briefly check low-speed rotation with airframe restrained. For conventional brushless reversal, power off and swap two phase wires. Prop handedness and orientation must produce forward thrust in the pusher installation.

Failsafes: prove the trigger

Prop removed; disconnect motor power for configuration tests

A switch labelled failsafe is not a link-loss test. With Mission Planner recording, power the assembled radio path, select a test mode and switch the transmitter off. Observe inputs, failsafe messages and resulting mode; restore link and observe recovery.

Radio loss

Enable `THR_FAILSAFE=1`. Determine whether RX stops frames, flags loss or continues sending preset/held channels. Held normal values may prevent detection. Configure a detectable loss condition and repeat the actual transmitter-off test. [13]

For low-throttle triggering, place `THR_FS_VALUE` between measured loss throttle and lowest valid throttle, with reliable separation. Example only: valid low 1000, loss 900, threshold 950 microseconds. If RX cannot provide a distinct condition, change the setup.

Candidate fixed-wing policy: short loss `CIRCLE` (`FS_SHORT_ACTN=1`), sustained loss `RTL` (`FS_LONG_ACTN=1`). Record both timeouts and verify firmware action labels. Home, return height, terrain clearance and available power must be valid before using this flight policy. [13]

Battery and navigation failures

Record `BATT_LOW_VOLT`, `BATT_CRT_VOLT`, `BATT_LOW_MAH`, `BATT_CRT_MAH`, `BATT_LOW_TIMER` and low/critical actions. Confirm units and action labels in installed firmware. Test using a suitable variable supply within FC limits or temporary threshold changes; restore, read back and save flight values afterward. Do not deeply discharge LiPo to trigger a test.

Ground RTL testing proves a requested mode change, not a safe return path. Test navigation-failure policies in simulation before controlled flight evaluation with an experienced pilot. Do not unplug live GPS cables or obstruct sensors near a spinning prop.

Record trigger, delay, action and recovery. Repeat from every intended flight mode. If a failsafe cannot be observed or explained, keep the flight gate closed.

Installation and release record

Complete on the actual aircraft

Keep the battery tray adjustable and restrained in all directions. Separate high-current wires from GPS/compass and RX antennas. Strain-relieve connectors and protect wires at foam/wood/carbon edges. Preserve access to USB, SD card, rail bridge and battery connector. Label detachable wing/tail leads.

Provisional setup balance station

Mark 522 mm aft of the CAD nose tip, or 22.2 mm aft of the front wing-joiner axis. This is the geometric quarter-MAC initial balancing reference, not an aerodynamic or flight-approved CG range. CAD mesh integration gives exposed wing area 304,451 mm², MAC 169.827 mm and quarter-MAC station 522.312 mm aft nose. Flight release still needs measured mass, control authority and experienced-pilot review.

xCG = sum(mass x position) / total mass

Battery adjustment: battery shift = total mass / battery mass x desired CG shift.

Use one nose datum, aft positive. Include wiring, glue and finish. Mark repeatable pack positions; verify assembled balance and lateral balance.

Record	Measured / approved value
Firmware / saved parameters	_____
5 V minimum / Vx minimum / servo peak current	_____
Voltage calibration / current reference	_____
RC loss trigger / short / long / recovery	_____
Battery low / critical / actions	_____
Total mass / battery mass / actual CG from nose	_____
Desired CG shift / required battery movement	_____
Control throws / propulsion test / flight CG review	_____

Flight gate: identified connectors; stable loaded supplies; correct control/stabilization direction; range and failsafe checks; valid sensors; structural/CG approval; qualified propulsion. Motor + 9045 three-blade + 40 A ESC compatibility remains unproved. A prop-off spin does not qualify it. Use a proper restrained fixture and experienced supervision; measure current, thrust, voltage and temperature.

References and handoff

Primary sources checked 19 September 2026

[1] Matek V2 specifications, layout and mapping

<https://www.mateksys.com/?portfolio=f405-wing-v2>

[2] ArduPilot F405-Wing (older details need V2 cross-check)

<https://ardupilot.org/plane/docs/common-matekf405-wing.html>

[3] ArduPilot Choosing Servo Functions

<https://ardupilot.org/plane/docs/servo-functions.html>

[4] ArduPilot board hardware definition (master)

https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-Wing/hwdef.dat

[5] FlySky FS-iA10B specification

<https://www.flysky-cn.com/ia10b-canshu>

[6] Matek ASPD-4525

<https://www.mateksys.com/?portfolio=aspd-4525>

[7] Matek ASPD-DLVR

<https://www.mateksys.com/?portfolio=aspd-dlvr>

[8] ArduPilot RC input mapping and reversal

<https://ardupilot.org/plane/docs/rc-throw-trim.html>

[9] ArduPilot V-tail / inverted V-tail setup

<https://ardupilot.org/plane/docs/guide-vtail-plane.html>

[10] ArduPilot Airspeed Calibration

<https://ardupilot.org/plane/docs/calibrating-an-airspeed-sensor.html>

[11] ArduPilot Airspeed Parameters Setup

<https://ardupilot.org/plane/docs/airspeed-parameters-setup.html>

[12] ArduPilot ESC Calibration

<https://ardupilot.org/plane/docs/common-esc-calibration.html>

[13] ArduPilot Plane Failsafe Function

<https://ardupilot.org/plane/docs/apms-failsafe-function.html>

[14] ArduPilot Ground Steering

<https://ardupilot.org/plane/docs/tuning-ground-steering-for-a-plane.html>

[15] Emax ECO II 2807 official drawing in product description

<https://emaxmodel.com/products/emax-eco-ii-series-2807-1300kv-1700kv-1500kv-brushless-motor-for-rc-drone-fpv-racing>

Continuity: output/pdf/MQ9_Hardware_and_CG_Notes.pdf, 19 September 2026. Exact ESC, servo, GPS and airspeed manuals remain required. No aircraft current, thrust, servo load, firmware readback or flight evidence was collected.

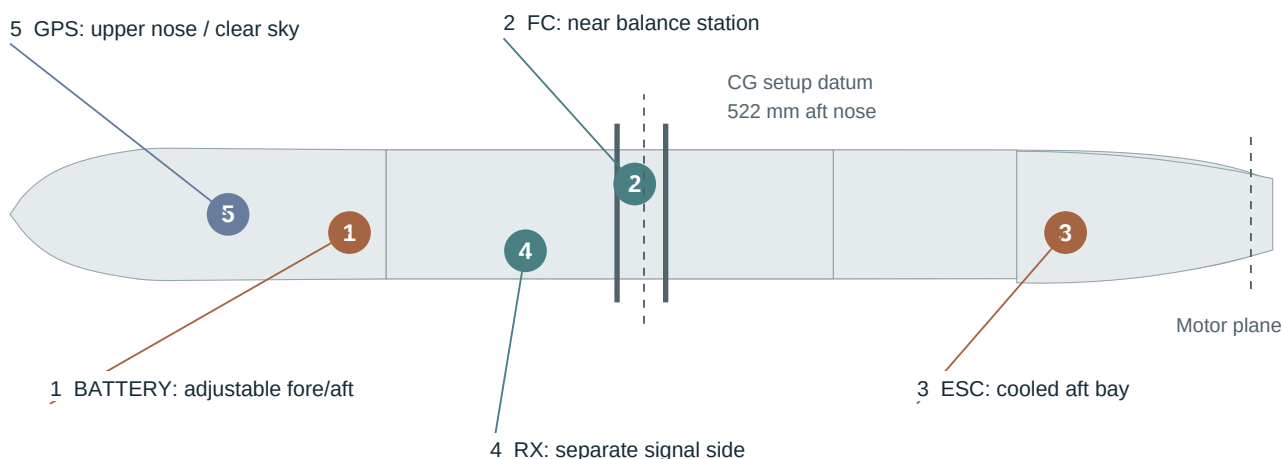
Where the electronics go

Proposed zones on the actual CAD body - not mounting-hole templates

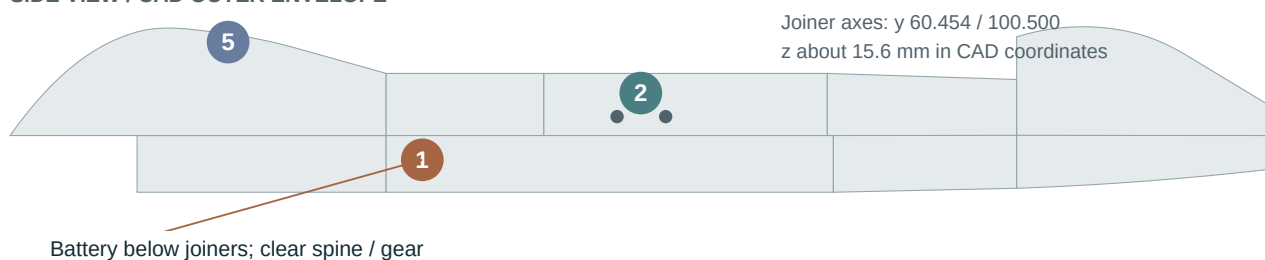
Place equipment with the bare structure open. These zones follow the assembled fuselage and wing-joiner geometry; they do not confirm battery fit. The GenX pack dimensions and its cable exit must be checked before a tray or hatch is made.

TOP VIEW / NOSE LEFT

PUSHER AFT



SIDE VIEW / CAD OUTER ENVELOPE

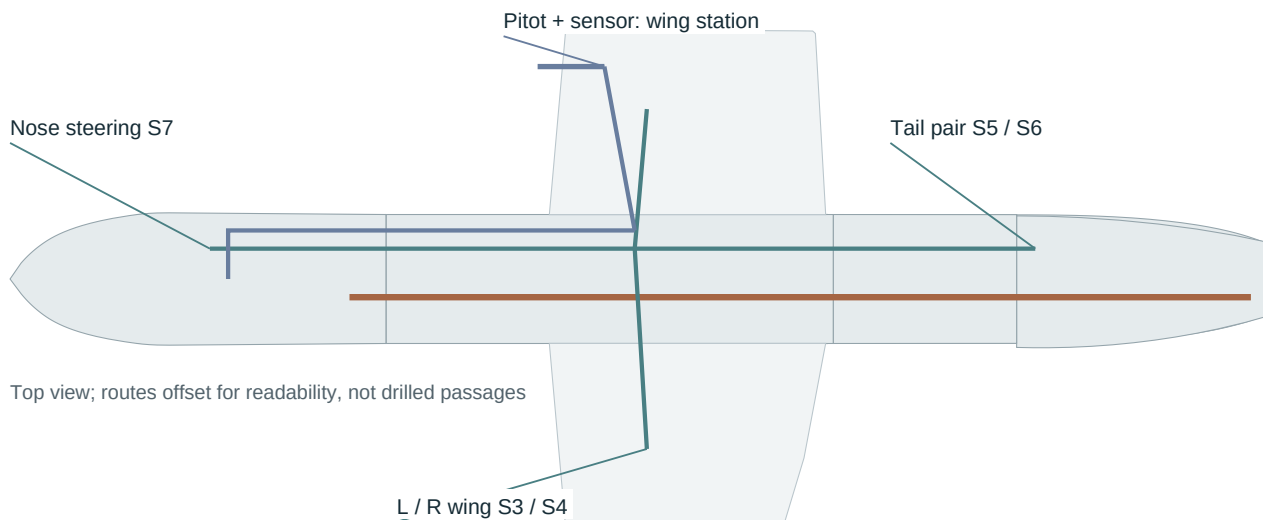


Zone	Mounting decision
1 Battery	Adjustable tray below the wing tubes, to one side of the centre wood spine as space permits. Clear nose/main gear and tube roof. Positive restraint in every direction; accessible connector. No pack dimensions or fitted tray claimed.
2 Flight controller	Accessible near the setup balance station on a firm mounting platform, using documented board grommets. Arrow forward; clear USB/SD access, level datum and wires that pull on the board.
3 / 4 / 5 ESC, RX, GPS	ESC aft with airflow and service access; RX on the signal side. Put M9 antenna upward with clear sky; if compass equipped, separate it from battery/current wiring and steel. Place RX active antenna ends clear of carbon and motor power. [16]

Wing pitot placement and harness routes are shown next. Colour marks identify functions; their drawn size and coordinates are not component footprints.

Harness routes and mounting

Keep removable joints accessible; confirm fit before closing skins



Harness	Route and service point
Brown: high current	Battery -> FC distribution -> ESC -> motor. Keep + and - together and away from signal/sensor bundles. Keep ESC input leads short; obtain the actual ESC maker cable-length/capacitor guidance before extending. No unsupported length limit is assumed.
Teal: control	Separate 3-wire S3/S4 wing leads, labelled L / R with signal, servo positive and ground. Retain keyed connectors at wing roots. Route S5/S6 to the tail and S7 to nose steering; secure leads clear of hinges, pushrods and gear.
Blue: navigation	GPS UART and optional compass to FC; airspeed I2C to a supported sensor near its pitot. Mount probe upstream of the wing leading edge in undisturbed flow, away from fuselage boundary layer and prop wash. Keep tubes short, unkinked and removable. Do not choose a fuselage-roof position just because it is convenient. [17]

Install in six passes

1. Mock up actual battery, FC, ESC and receiver; mark hatch access. 2. Confirm tray travel around spine, gear and joiners. 3. Install mounts, then route power on one side and signal wiring separately. 4. Add labelled wing/tail service connectors with strain relief. 5. Check pack restraint, cooling, full linkage travel and antenna exposure with hatches shut. 6. Repeat rail-load, radio-range, sensor, control-direction and balance checks before closing the build.

CAD basis: assembled meshes A06/A08/A12/A15/A17/A20/A42/A56/A59; y increases aft. Nose y -439.65, motor mounting plane y 582.65, quarter-MAC setup station y 82.662 mm. Zones are engineering placement proposals, not proven fits.

[16] ArduPilot GPS/compass mounting

[17] ArduPilot Pitot Tube Considerations

Detailed parts list

Quantities are for one complete airframe. Card forming tools and the print coupon are not installed aircraft parts. IDs match the drawings, files and assembly maps.

ID	Part	Qty	Material / process
A01	Rear joiner support	2	5.000 mm plywood / Plywood cut
A04	Aft fuselage former	1	5.000 mm plywood / Plywood cut
A05	Front wing spar	2	5.000 mm plywood / Plywood cut
A07	Rear wing spar	2	5.000 mm plywood / Plywood cut
A10	Outer aileron end plate	2	3.000 mm plywood / Plywood cut
A13	Forward longitudinal former	1	5.000 mm plywood / Plywood cut
A16	Middle fuselage former	1	5.000 mm plywood / Plywood cut
A18	Tail bulkhead	1	2.500 mm plywood / Plywood cut
A24	Middle wing rib	2	5.000 mm plywood / Plywood cut
A25	Front gear bulkhead	1	5.000 mm plywood / Plywood cut
A26	Motor mount	1	5.000 mm plywood / Plywood cut
A34	Forward fuselage former	1	5.000 mm plywood / Plywood cut
A35	Inner wing rib	2	5.000 mm plywood / Plywood cut
A43	Aft longitudinal former	1	5.000 mm plywood / Plywood cut
A44	Outer wing rib	2	4.990 mm plywood / Plywood cut
A47	Main gear base	1	5.000 mm plywood / Plywood cut
A48	Rear gear bulkhead	1	5.000 mm plywood / Plywood cut
A53	Inner aileron end plate	2	3.000 mm plywood / Plywood cut
A54	Central forward former	1	5.000 mm plywood / Plywood cut
A62F	Front nose spine segment	1	5.000 mm plywood / Plywood cut
A62R	Rear spine segment	1	5.000 mm plywood / Plywood cut
A63	Nose cheek doublers (A63/A64)	2	2.500 mm plywood / Plywood cut
A06	Centre fairing	1	LW-PLA preferred; PLA+ mass must be budgeted / 3D print
A09	Nose steering bearing	1	PLA+ or PETG; functional coupon required / 3D print
A11	Left aileron	1	LW-PLA preferred; PLA+ mass must be budgeted / 3D print

ID	Part	Qty	Material / process
A12	Tail fairing	1	LW-PLA preferred; PLA+ mass must be budgeted / 3D print
A21	Right aileron	1	LW-PLA preferred; PLA+ mass must be budgeted / 3D print
A23	Left aileron hinge	1	LW-PLA preferred; PLA+ mass must be budgeted / 3D print
A36	Right aileron hinge	1	LW-PLA preferred; PLA+ mass must be budgeted / 3D print
A37	Nose gear mount	1	PLA+ or PETG; functional coupon required / 3D print
A45	Right wing root	1	PLA+ or PETG; functional coupon required / 3D print
A55	Left wing root	1	PLA+ or PETG; functional coupon required / 3D print
A59	Nose shell	1	LW-PLA preferred; PLA+ mass must be budgeted / 3D print
F01	Left inner wing	1	5 mm FliteBoard Pro / Knife cut and form
F02	Right inner wing	1	5 mm FliteBoard Pro / Knife cut and form
F03	Right outer wing	1	5 mm FliteBoard Pro / Knife cut and form
F04	Left outer wing	1	5 mm FliteBoard Pro / Knife cut and form
F05	Right fixed tail - main	1	5 mm FliteBoard Pro / Knife cut and form
F06	Left fixed tail - main	1	5 mm FliteBoard Pro / Knife cut and form
F07	Right fixed tail - root collar	1	5 mm FliteBoard Pro / Knife cut and form
F08	Left fixed tail - root collar	1	5 mm FliteBoard Pro / Knife cut and form
F09	Forward lower fuselage - left curve	1	5 mm FliteBoard Pro / Knife cut and form
F10	Forward lower fuselage - right curve	1	5 mm FliteBoard Pro / Knife cut and form
F11	Middle lower fuselage - right curve	1	5 mm FliteBoard Pro / Knife cut and form
F12	Middle lower fuselage - left curve	1	5 mm FliteBoard Pro / Knife cut and form
F13	Forward lower fuselage - left lip	1	5 mm FliteBoard Pro / Knife cut and form
F14	Forward lower fuselage - bottom strip	1	5 mm FliteBoard Pro / Knife cut and form
F15	Forward lower fuselage - right lip	1	5 mm FliteBoard Pro / Knife cut and form
F16	Middle lower fuselage - left lip	1	5 mm FliteBoard Pro / Knife cut and form
F17	Middle lower fuselage - right lip	1	5 mm FliteBoard Pro / Knife cut and form
F18	Middle lower fuselage - bottom strip	1	5 mm FliteBoard Pro / Knife cut and form

ID	Part	Qty	Material / process
F19	Aft lower fuselage - left lip	1	5 mm FliteBoard Pro / Knife cut and form
F20	Aft lower fuselage - left curve	1	5 mm FliteBoard Pro / Knife cut and form
F21	Aft lower fuselage - right curve	1	5 mm FliteBoard Pro / Knife cut and form
F22	Aft lower fuselage - right lip	1	5 mm FliteBoard Pro / Knife cut and form
F23	Aft upper fuselage - left	1	5 mm FliteBoard Pro / Knife cut and form
F24	Aft upper fuselage - right	1	5 mm FliteBoard Pro / Knife cut and form
F25	Forward upper fuselage - right	1	5 mm FliteBoard Pro / Knife cut and form
F26	Forward upper fuselage - left	1	5 mm FliteBoard Pro / Knife cut and form
F27	Rear underside band 1 - forward	1	5 mm FliteBoard Pro / Knife cut and form
F28	Rear underside band 2	1	5 mm FliteBoard Pro / Knife cut and form
F29	Rear underside band 3	1	5 mm FliteBoard Pro / Knife cut and form
F30	Rear underside band 4 - aft	1	5 mm FliteBoard Pro / Knife cut and form
F31	Right tail control laminate blank	2	5 mm FliteBoard Pro / Knife cut and form
F32	Left tail control laminate blank	2	5 mm FliteBoard Pro / Knife cut and form
F33	Aft lower fuselage - bottom strip	1	5 mm FliteBoard Pro / Knife cut and form
T01	T01 front joiner	1	Carbon OD8/ID6 mm / Purchase / cut tube
T02	T02 right front sleeve	1	Carbon OD10/ID8.2 mm / Purchase / cut tube
T03	T03 left front sleeve	1	Carbon OD10/ID8.2 mm / Purchase / cut tube
T04	T04 rear joiner	1	Carbon OD8/ID6 mm / Purchase / cut tube
H01	Main formed landing gear	1	12.7 x3.175 mm suitable formed metal / Hardware
H02	Nose wire extended	1	3 mm spring steel wire / Hardware
H03	Wheels	3	50.8 mm diameter,3 mm bore / Hardware
H04	Servos	5	9 g metal gear; actual model fit/voltage check / Hardware
H05	Control horns	4	Purchased lightweight RC horns / Hardware
H06	Nose steering arm	1	3 mm clamping arm / Hardware
H07	Pushrod and clevis sets	5	Matched RC linkage / Hardware
H08	Aileron pins	2	3 mm smooth rod; start285 mm / Hardware
H09	Fastener pack	1	See airframe guide page3 and hardware instructions / Hardware

Hardware and forming notes

HARDWARE AND FIT REFERENCES - millimetres

A02 main landing gear: formed strip reference, not a plastic print and not a laser flat blank. Source section is 12.7 mm fore-aft width by 3.175 mm strip thickness. Overall stance approx 285.5 mm; main axle centres $x = \pm 151$ mm with wheels, axle height $z = -135.85$. Use a suitable formed model-aircraft metal gear or commission this geometry. The metalwork shop must select a bendable alloy/temper and finite bend radii appropriate to it, then preserve attachment and axle stations. Source CAD has ideal sharp bends; do not interpret these as zero-radius physical bends. Start with a strip longer than developed centreline, form to the STEP/template and finish ends. Shop calculates bend deductions for its tooling. Two new 3.2 mm attachment holes $x = \pm 22.8222, y = 136.35$ align A47; fit 2xM3 through-bolts, large washers and locking nuts. Suitable small-aircraft landing gear requires physical load/deflection testing for completed aircraft mass.

A46 nose wire: 3 mm spring/piano-wire reference, with upper shaft extended to $z = +1$ mm so it reaches the steering bearing/arm. Preserve the source bends and axle line. Cut to the supplied formed STEP using a bending jig; final upper end can be trimmed only after bearing, shaft collar and steering arm are installed. Wire must rotate freely in finished 3.2 mm bearing bores without lateral play that permits wheel shimmy. Do not 3D print it. Final upper length depends on actual clamp/arm.

Wheels: 3 purchased lightweight 50.8 mm diameter, approximately 16.5 mm wide, 3 mm bore. Two main axle bolts nominal M3x25 mm plus spacers/washers/locking nuts; nose wheel uses wire axle with 3 mm collars. Fit freely rotating wheels without clamping hubs.

Controls: 2x3 mm smooth hinge rods, start 285 mm long, final trim flush to outer end retainers with approximately 1 mm axial freedom. 4 purchased lightweight horns for 2 ailerons + 2 ruddervators; 1 clamping 3 mm nose-steering arm. 5 pushrods with matching clevises/ball links, cut only after neutral servo and surface positions are set. Do not use glue alone to retain hinge pins or wing joiners.

Root wing retention: six M2 through-bolts, start 20 mm length, washers and locking nuts in three aligned root attachment holes per wing. Finish printed/wood holes to the actual fastener clearance. Set lengths to avoid moving linkages. These root fasteners retain removable wings; do not glue the sliding carbon joiner into both wings.

Motor: four M3 bolts through revised A26, 19 mm PCD. Select length to pass 5 mm plywood + washer and engage motor threads without reaching windings. Actual screw engagement depth must be measured; do not blindly reuse a long screw. Manufacturer drawing is supplied.

The 80x35x4 mm print coupon has bores 2.0, 2.2, 3.0, 3.2, 8.0, 8.2, 10.2 mm. Use it to establish printer/material fit; it is not an aircraft part. Size must remain 100%. Verify outside dimensions and fit using intended rods/bolts before printing structural brackets.

Nose bearing A09 to mount A37: four M3 machine screws with washers and locking nuts, start 16-20 mm long and select after measuring the actual stack. The shaft-through portion of the mount is 7 mm and the bearing foot about 3 mm, plus washer/nut stack. Use the printed underside access recesses; do not trap a loose nut where it cannot be inspected. Finish the bearing clearance holes to the actual screw and verify no bolt can touch the rotating nose shaft.

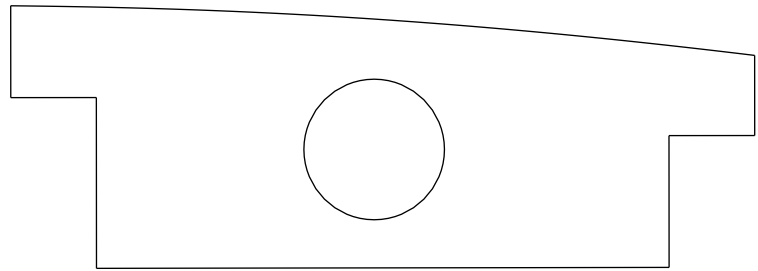
MQ9 manufacturing profiles - 22 profiles / 31 pieces

Preview only; DXF units mm, 1:1 native curves. Qty includes inferred mirrored wing members. Declared bore revisions and prismatic blanks; CUT layer only.

A01 MQ9 Assembled v28

5.000 mm | 43.40 x 15.32 mm | Qty 2 | 1 internal loops

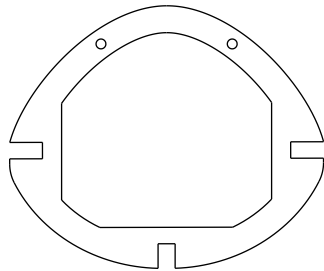
REVISED MANUFACTURING PROFILE



A04 Aft_Fuselage_Hoop v1

5.000 mm | 95.65 x 80.03 mm | Qty 1 | 3 internal loops

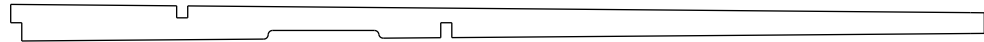
REVISED MANUFACTURING PROFILE



A05 Front_Spar v1

5.000 mm | 440.00 x 16.54 mm | Qty 2 | 0 internal loops

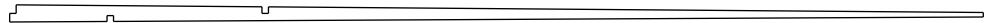
SOURCE CONTOUR



A07 Rear_Spar v1

5.000 mm | 750.00 x 13.19 mm | Qty 2 | 0 internal loops

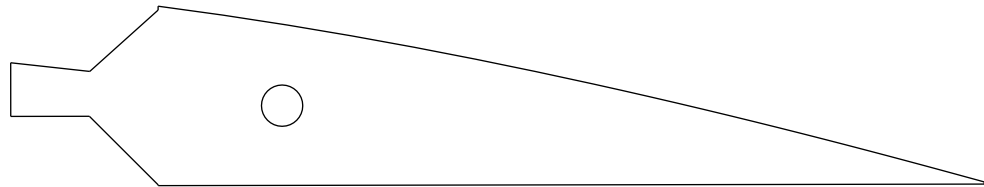
SOURCE CONTOUR



A10 Outer_Aileron_Plate v1

3.000 mm | 70.72 x 13.03 mm | Qty 2 | 1 internal loops

PRISMATIC BLANK - FINISH EDGE



A13 Front_Lateral_Hoop v1

5.000 mm | 95.64 x 550.66 mm | Qty 1 | 10 internal loops

SOURCE CONTOUR



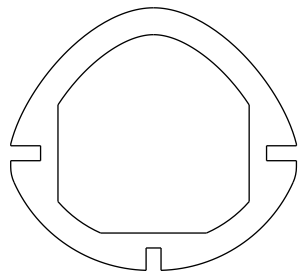
MQ9 manufacturing profiles - 22 profiles / 31 pieces

Preview only; DXF units mm, 1:1 native curves. Qty includes inferred mirrored wing members. Declared bore revisions and prismatic blanks; CUT layer only.

A16 Fwd_Fuselage_Hoop v1 (2)

5.000 mm | 95.64 x 87.90 mm | Qty 1 | 1 internal loops

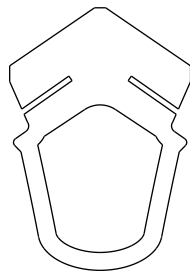
SOURCE CONTOUR



A18 MQ9 Assembled v28

2.500 mm | 81.00 x 117.59 mm | Qty 1 | 1 internal loops

SOURCE CONTOUR



A24 Middle_Rib v1

5.000 mm | 153.89 x 13.56 mm | Qty 2 | 2 internal loops

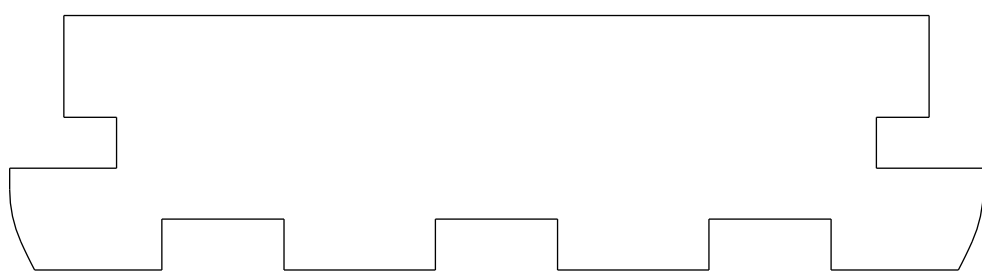
REVISED MANUFACTURING PROFILE



A25 MQ9 Assembled v28

5.000 mm | 95.64 x 25.00 mm | Qty 1 | 0 internal loops

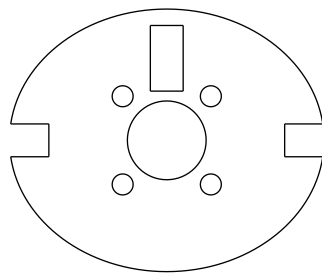
SOURCE CONTOUR



A26 Notor_Mount v1

5.000 mm | 47.55 x 40.00 mm | Qty 1 | 6 internal loops

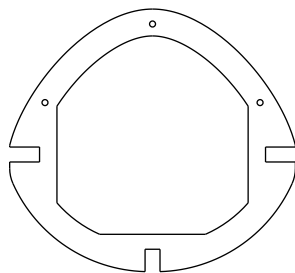
REVISED MANUFACTURING PROFILE



A34 Fwd_Fuselage_Hoop v1 (1)

5.000 mm | 95.64 x 87.90 mm | Qty 1 | 4 internal loops

SOURCE CONTOUR



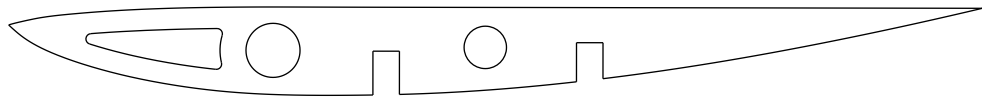
MQ9 manufacturing profiles - 22 profiles / 31 pieces

Preview only; DXF units mm, 1:1 native curves. Qty includes inferred mirrored wing members. Declared bore revisions and prismatic blanks; CUT layer only.

A35 Inner_Rib v1

5.000 mm | 183.66 x 16.67 mm | Qty 2 | 3 internal loops

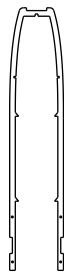
REVISED MANUFACTURING PROFILE



A43 Aft_Lateral_Hoop v1

5.000 mm | 95.64 x 467.00 mm | Qty 1 | 4 internal loops

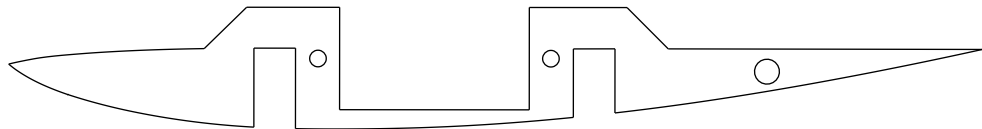
SOURCE CONTOUR



A44 Outer_Rib v1

4.990 mm | 117.03 x 14.63 mm | Qty 2 | 3 internal loops

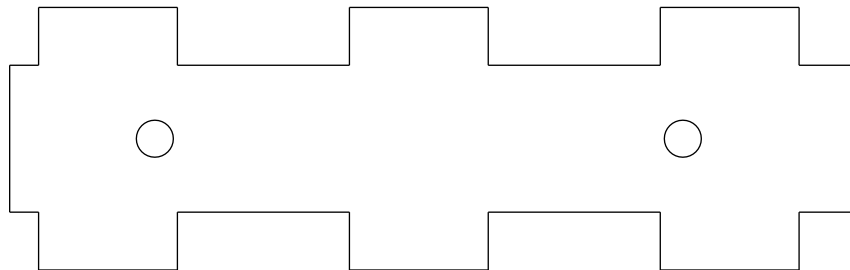
PRISMATIC BLANK - FINISH EDGE



A47 MQ9 Assembled v28

5.000 mm | 72.74 x 22.70 mm | Qty 1 | 2 internal loops

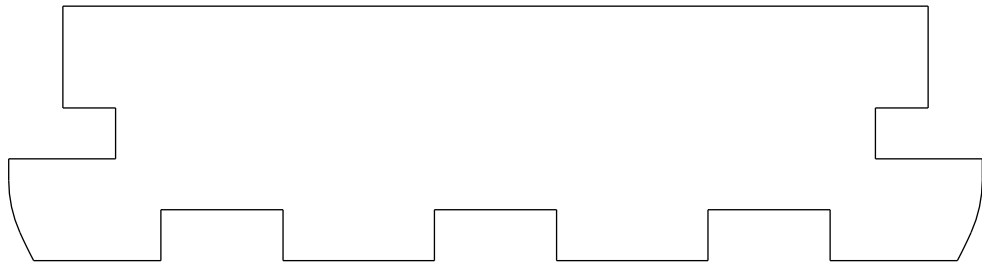
REVISED MANUFACTURING PROFILE



A48 MQ9 Assembled v28

5.000 mm | 95.64 x 25.00 mm | Qty 1 | 0 internal loops

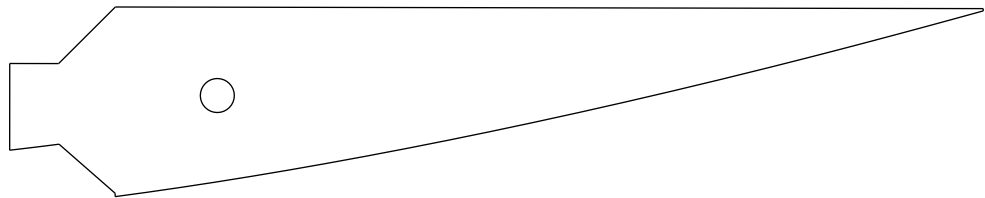
SOURCE CONTOUR



A53 Inner_Aileron_Plate v1

3.000 mm | 85.87 x 16.73 mm | Qty 2 | 1 internal loops

SOURCE CONTOUR



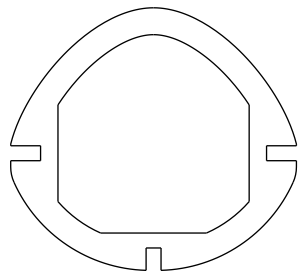
MQ9 manufacturing profiles - 22 profiles / 31 pieces

Preview only; DXF units mm, 1:1 native curves. Qty includes inferred mirrored wing members. Declared bore revisions and prismatic blanks; CUT layer only.

A54 Fwd_Fuselage_Hoop v1

5.000 mm | 95.64 x 87.90 mm | Qty 1 | 1 internal loops

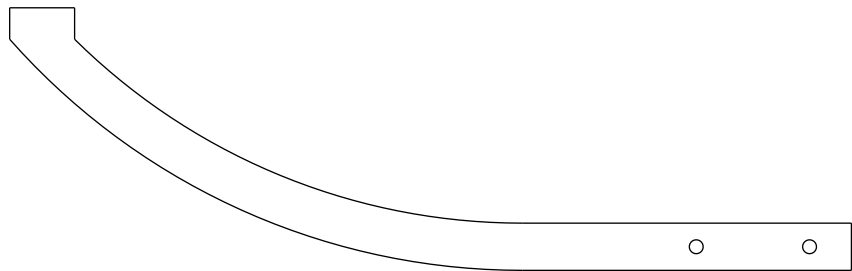
SOURCE CONTOUR



A62F Unified spine forward segment

5.000 mm | 133.72 x 41.71 mm | Qty 1 | 2 internal loops

REVISED MANUFACTURING PROFILE



A62R Unified spine rear segment

5.000 mm | 887.62 x 41.71 mm | Qty 1 | 2 internal loops

REVISED MANUFACTURING PROFILE



A63 Nose gear continuous side doubler pair (A63/A64)

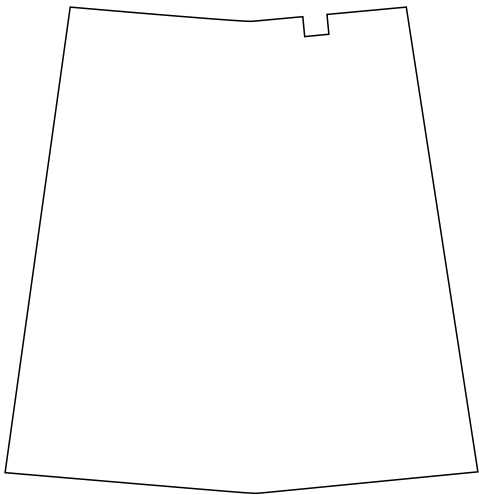
2.500 mm | 72.00 x 10.70 mm | Qty 2 | 4 internal loops

REVISED MANUFACTURING PROFILE

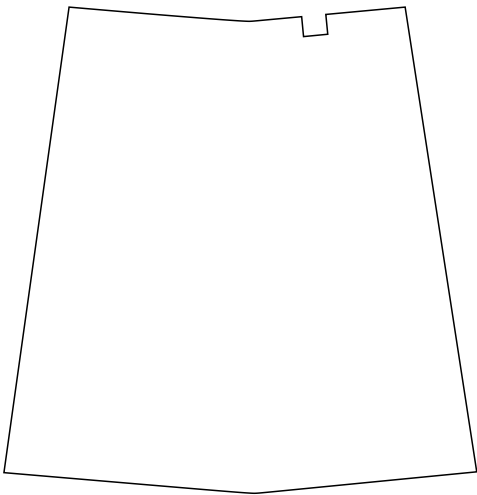


MQ9 | developed foam patterns | overview only

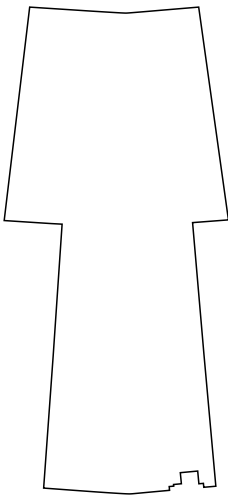
Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



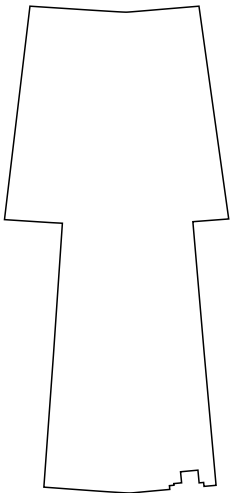
F01 Developed skin
wing_inner_A
443.1 x 455.6 mm | max strain 0.341%



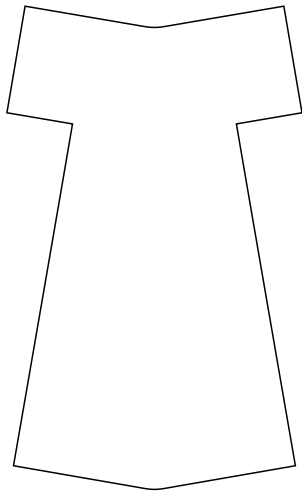
F02 Developed skin
wing_inner_B
443.1 x 455.6 mm | max strain 0.341%



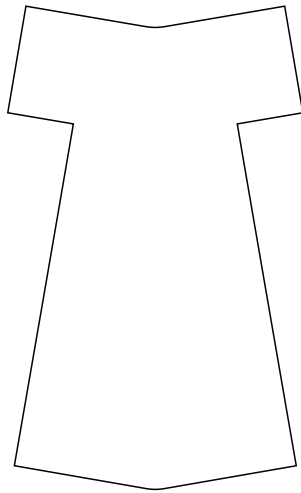
F03 Developed skin
wing_outer_A
237.1 x 514.7 mm | max strain 0.357%



F04 Developed skin
wing_outer_B
237.1 x 514.7 mm | max strain 0.366%



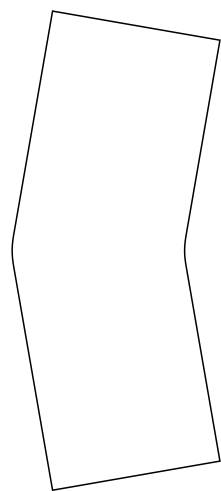
F05 Developed skin
tail_A_rootcollar1
177.4 x 290.5 mm | max strain 0.278%



F06 Developed skin
tail_B_rootcollar1
177.4 x 290.5 mm | max strain 0.278%

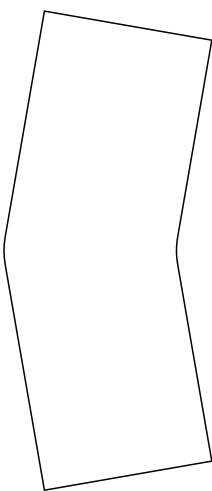
MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F07 Developed skin

tail_A_rootcollar2
83.6 x 192.9 mm | max strain 0.174%



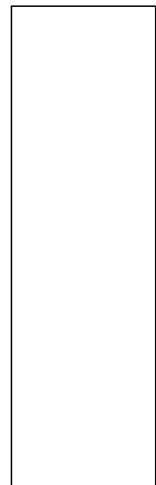
F08 Developed skin

tail_B_rootcollar2
83.6 x 192.9 mm | max strain 0.174%



F09 Developed skin

fuselage_forward_A
61.6 x 205.0 mm | max strain 0.023%



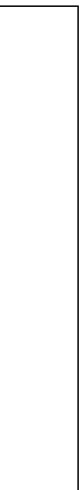
F10 Developed skin

fuselage_forward_B
61.6 x 205.0 mm | max strain 0.023%



F11 Developed skin

fuselage_middle_A
61.6 x 368.7 mm | max strain 0.023%



F12 Developed skin

fuselage_middle_B
61.6 x 368.6 mm | max strain 0.023%

MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F13 Developed skin

fuselage_src3_face8
14.4 x 205.0 mm | max strain 0.018%



F14 Developed skin

fuselage_src3_face10
5.3 x 205.0 mm | max strain 0.000%



F15 Developed skin

fuselage_src3_face12
14.4 x 205.0 mm | max strain 0.019%



F16 Developed skin

fuselage_src21_face2
14.4 x 368.7 mm | max strain 0.018%



F17 Developed skin

fuselage_src21_face16
14.4 x 368.7 mm | max strain 0.019%

F18 Developed skin

fuselage_src21_face18
5.3 x 368.7 mm | max strain 0.000%

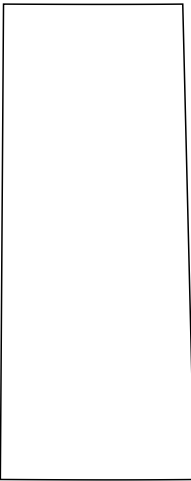
MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F19 Developed skin

fuselage_src10_face16
14.6 x 151.0 mm | max strain 0.000%



F20 Developed skin

fuselage_src10_face17
61.6 x 151.1 mm | max strain 0.037%



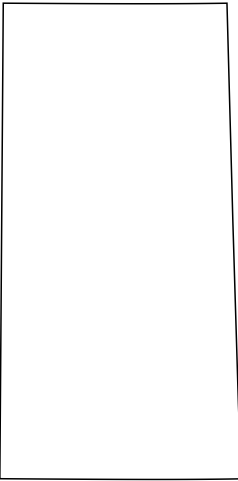
F21 Developed skin

fuselage_src10_face19
61.8 x 151.4 mm | max strain 0.035%



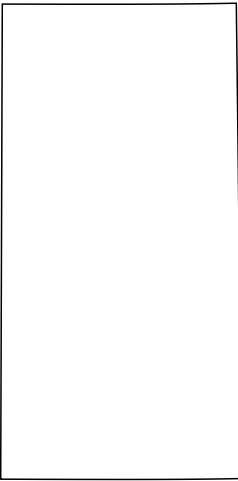
F22 Developed skin

fuselage_src10_face20
14.8 x 151.0 mm | max strain 0.000%



F23 Developed skin

fuselage_src22_face11
79.3 x 156.2 mm | max strain 0.030%

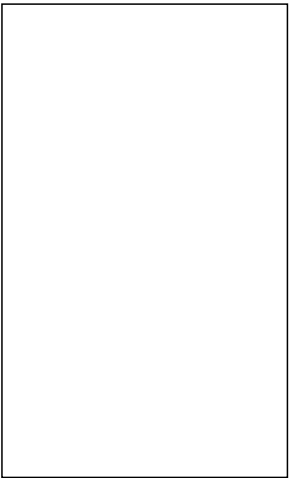


F24 Developed skin

fuselage_src22_face13
78.8 x 156.4 mm | max strain 0.037%

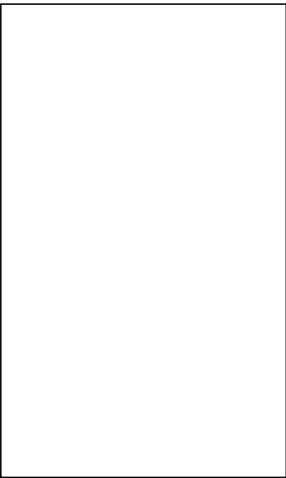
MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



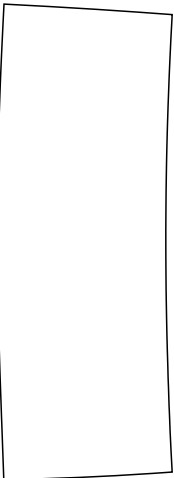
F25 Developed skin

fuselage_src25_face6
78.4 x 130.0 mm | max strain 0.000%



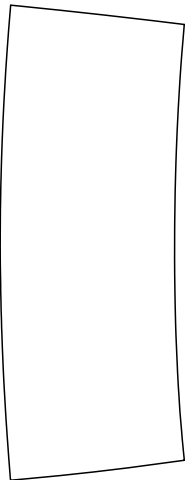
F26 Developed skin

fuselage_src25_face8
78.4 x 130.0 mm | max strain 0.000%



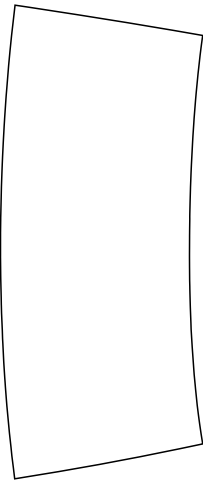
F27 Developed skin

fuselage_src18_face14_axis1_strip1of4
55.2 x 150.2 mm | max strain 0.287%



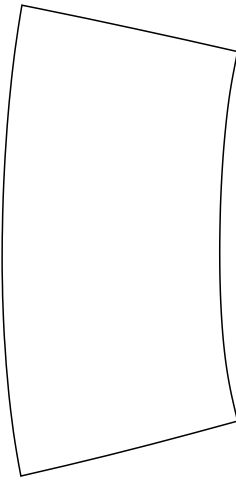
F28 Developed skin

fuselage_src18_face14_axis1_strip2of4
55.9 x 144.4 mm | max strain 0.295%



F29 Developed skin

fuselage_src18_face14_axis1_strip3of4
56.5 x 132.3 mm | max strain 0.336%



F30 Developed skin

fuselage_src18_face14_axis1_strip4of4
57.0 x 113.9 mm | max strain 0.321%

MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F31 Control blank

tail_control_blank_A
45.0 x 275.0 mm | max strain 0.000%



F32 Control blank

tail_control_blank_B
45.0 x 275.0 mm | max strain 0.000%

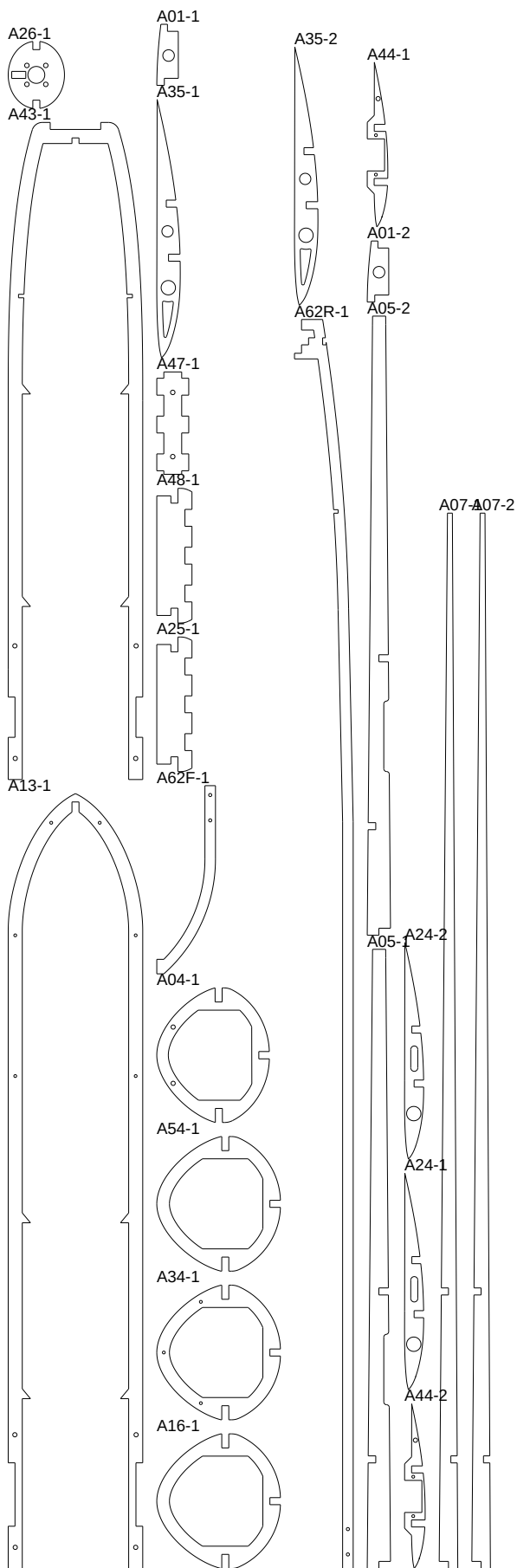


F33 Developed skin

fuselage_src10_face18
5.3 x 151.1 mm | max strain 0.000%

5mm stock: 24 pieces

Reference sheet1220x2440mm. Compact work area 362.8x1117.4mm. Grain +Y.



3mm stock: 4 pieces

Reference sheet1220x2440mm. Compact work area 36.7x363.2mm. Grain +Y.

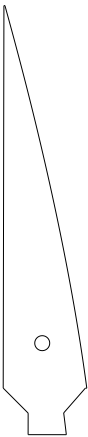
A10-2



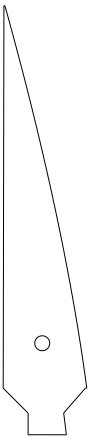
A10-1



A53-2



A53-1



2.5mm stock: 3 pieces

Reference sheet1220x2440mm. Compact work area 101.0x301.6mm. Grain +Y.

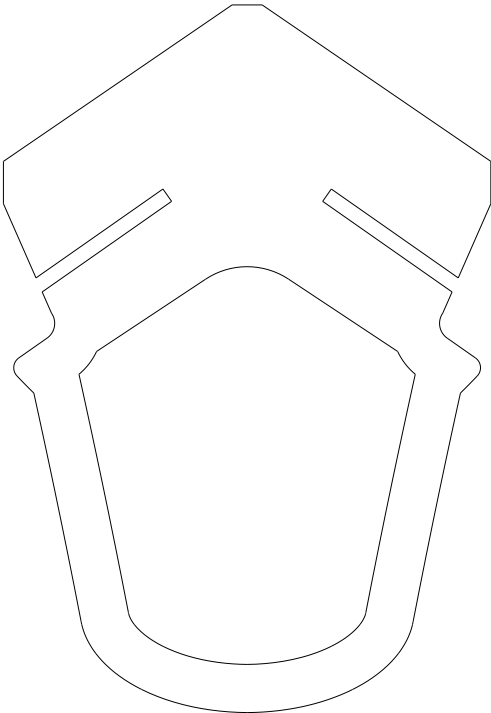
A63-2



A63-1

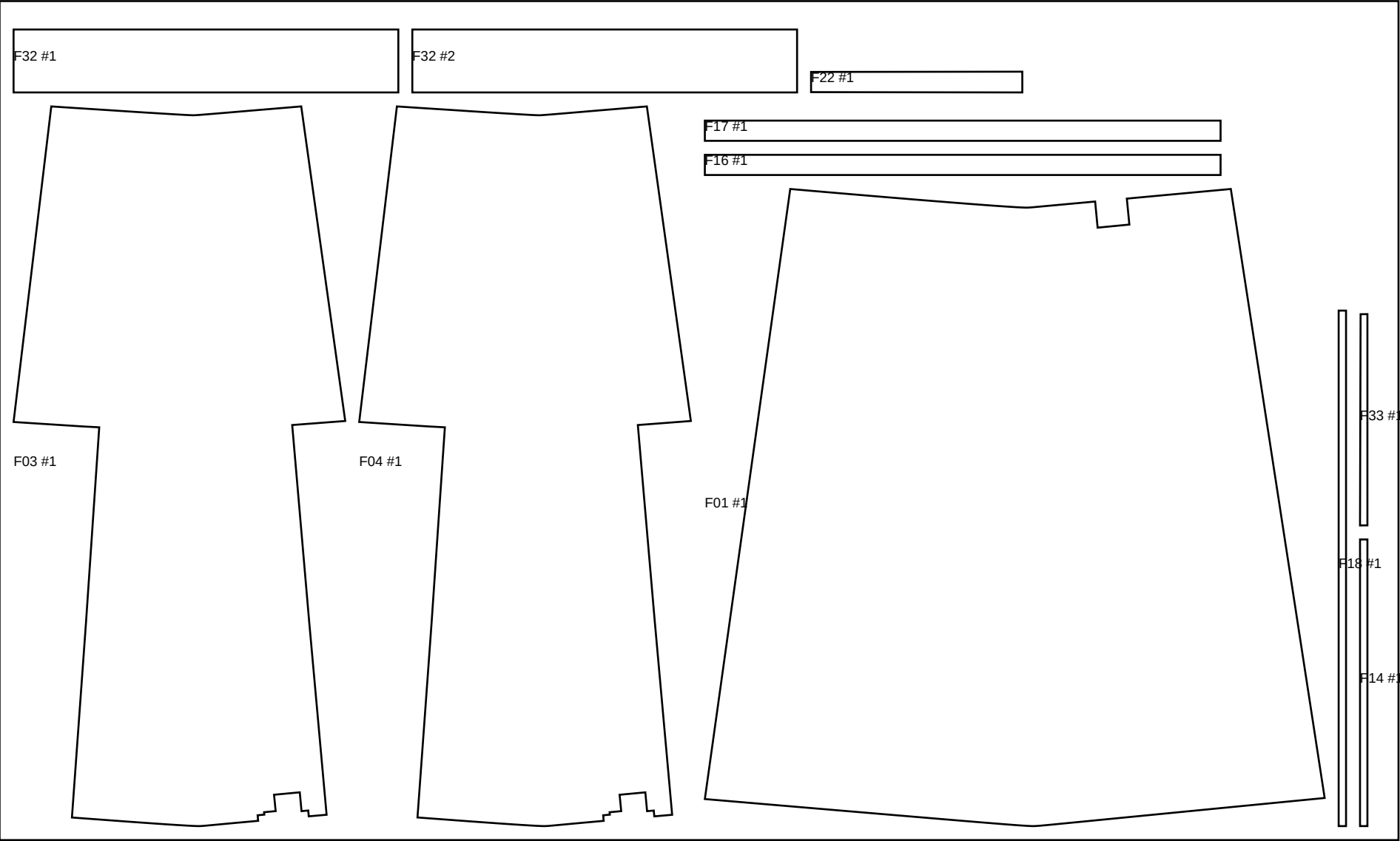


A18-1



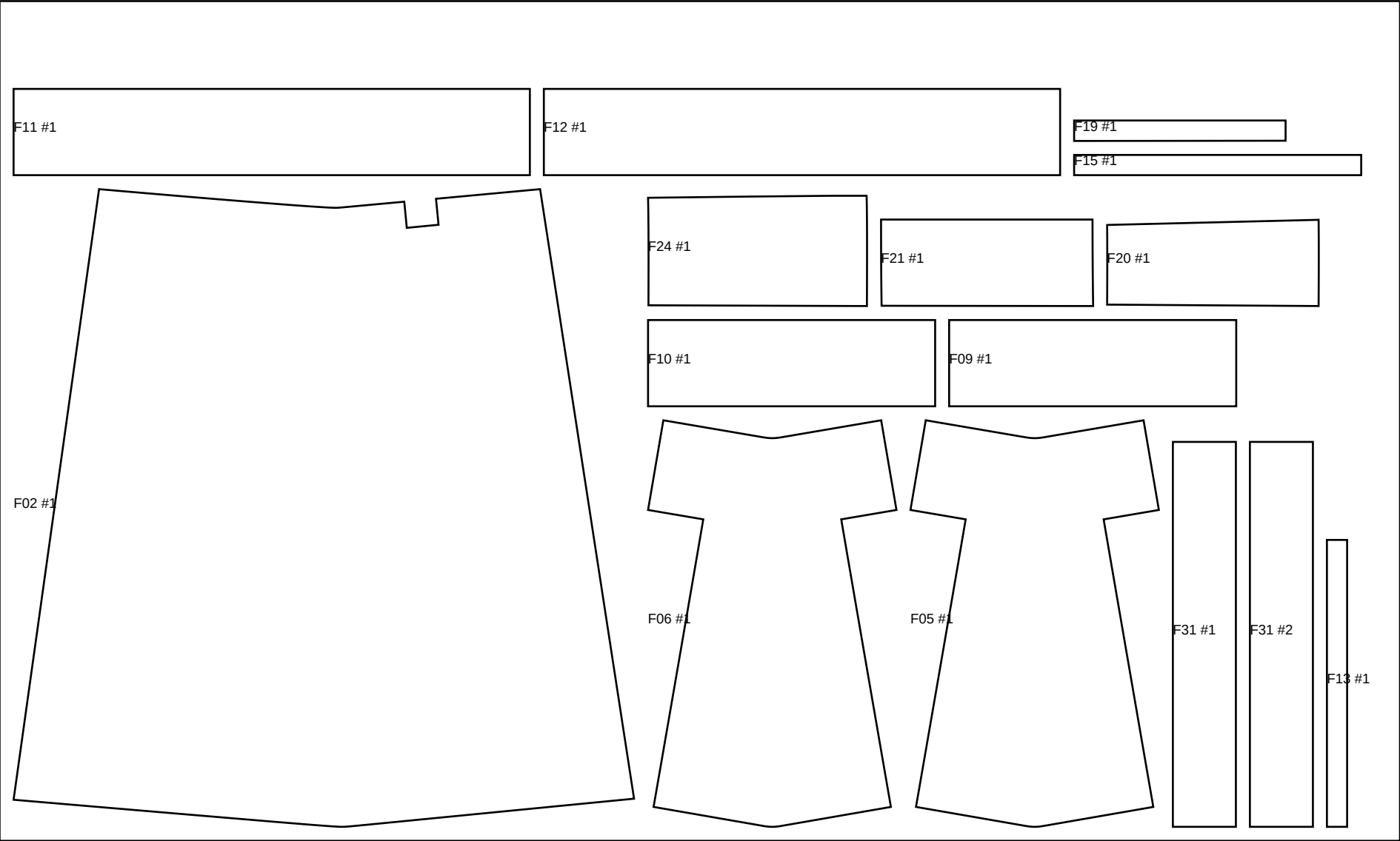
Foam sheet 1 / 1000 x 600 x 5 mm

Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



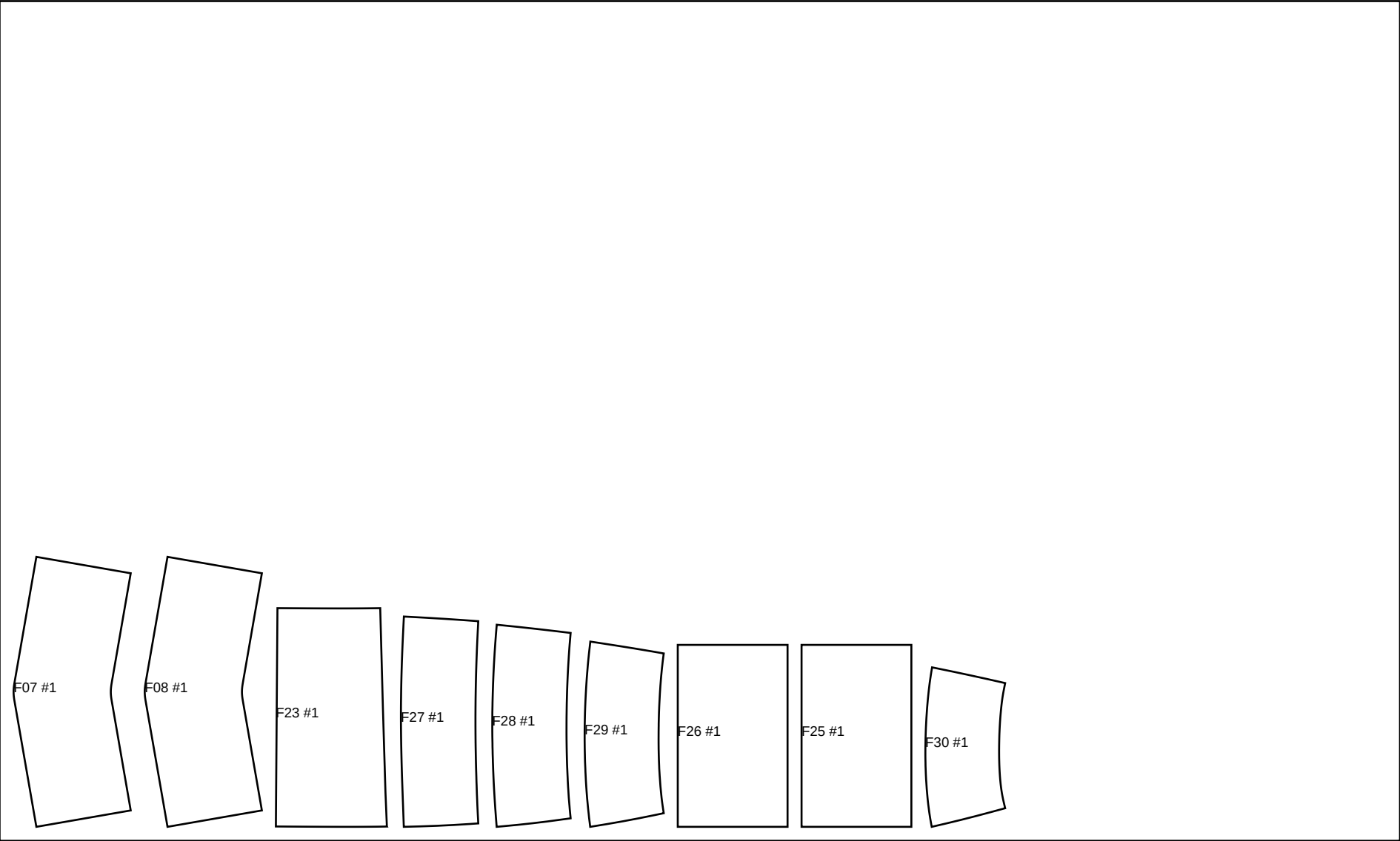
Foam sheet 2 / 1000 x 600 x 5 mm

Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



Foam sheet 3 / 1000 x 600 x 5 mm

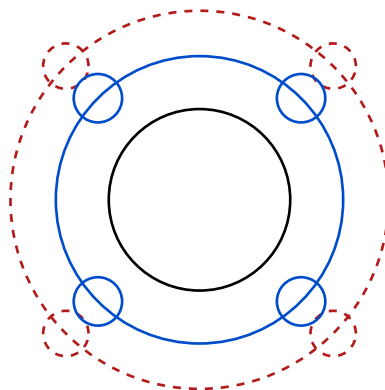
Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



A26 motor mount - declared drilling correction

Source: 4x3mm on 25mm PCD. Revised: 4x3.2mm on 19mm PCD.

Physical Emax motor check required before drilling; drawing below at 2:1.



Red dashed: obsolete 25mm PCD - DO NOT DRILL

Blue: 19mm PCD - four 3.2mm holes

Central 12mm hole unchanged. Minimum radial wood web 1.9mm; use washers.

Verification summary

MQ9 prototype fabrication package / Revision 1 / 19 September 2026

Check	Result and scope
Source inventory	62 original assembly occurrences retained for traceability; three tiny slivers excluded from fabrication. Assembly placements recovered independently of separated-part positions.
Plywood	22 profiles /31 pieces. Missing opposite-wing parts added. Native millimetre DXFs audit clean. Revised plate solids and source differences are recorded; revised blanks are not falsely labelled identical to tapered originals.
Plywood sheets / PDF	Three stock layouts; no bounding-box overlap.100 x 20 mm calibration verified from PDF vectors. Optimized plywood paper set uses 7 pages. Paired registration discrepancy at most 0.0000371 mm; sampled curve-scale discrepancy at most 0.0000717 mm.
Foam development	33 patterns /35 blanks. Maximum sampled mesh-edge strain 0.3664%; no reversed triangles or self-crossing boundaries. Absolute 0.2 mm mesh, checked against exact STEP surfaces. Maximum sampled normal departure 0.19971 mm. This is numerical geometry, not a material test.
Foam seams / sheets	Largest checked introduced-seam length mismatch 0.02667 mm. Three 1000 x 600 mm sheet layouts fit 35 blanks with no bounding-box overlap.Optimized foam paper set uses 23 pages including index/calibration. True-size coordinates and paired registration were rechecked; single-page-fit patterns remain whole.
Aircraft print meshes	11 meshes plus one separate test coupon. All aircraft meshes closed and consistently wound; manifold library reports NoError. Absolute export tolerance 0.015 mm. Maximum CAD/mesh volume discrepancy 0.05194%. Adaptive integration with BSpline spans avoids earlier misleading volumes.
Changed mating geometry	Zero computed volumetric overlap at extended nosewire/bearing, wire/mount, wire/spine, frontjoiner/sleeve, joiner/root, sleeve/middle rib and revised tailfairing/bulkhead. Typical nominal tube radial clearance 0.10 mm. Other local foam contact is explicitly handled by forming/relief instructions.

What changed from the source

1. Added the missing mirrored wing ribs, spars, end plates and rear joiner support; quantities refer to a full aircraft.
2. Replaced the contradictory front tube references with a centred 8/6 mm joiner and paired 10/8.2 mm sleeves. Recentred the rear 8/6 mm tube; omitted the redundant short source stub.
3. Moved the middle-rib sleeve bore onto the joiner axis and set 10.2 mm sleeve holes. Set 8.2 mm root/rear bores; physical tube tolerances remain a matched-fit shop check.
4. Combined overlapping centre spines. Split that spine deliberately at the nose shaft and added two 2.5 mm cheek plates, four M2 through-bolts and matching printed mount pockets. Extended the nosewire through its bearing/steering arm.
5. Changed motor holes from 25 mm to 19 mm pitch circle for the verified Emax drawing. Added/enlarged main-gear through-bolt holes for M3 fasteners.
6. Prepared prismatic wood blanks where the source had small side tapers. Sanding references and source departures are declared. Relieved the aft-hoop receiving notch without severing the spine.
7. Cleared the printed tailfaring bulkhead pocket and nose steering bore. Re-exported all selected print meshes with absolute, rather than relative, deflection.
8. Developed curved foam surfaces using a preserved outside-paper process for 5 mm FliteBoard. Tail controls are two-layer 10 mm blanks. Recomputed patterns on a finer absolute mesh and provided seam/relief instructions.

Operations still required at the workshop

Measure stock and run slot/print/foam coupons. Dry-fit all neighbours together. Finish prescribed plywood edges and foam inner-core relief; cut gear exits from actual installed geometry. Test printed moving fits and fastener stacks. Form metal gear using suitable material and finite tooling bend radii. Check structural joints and gear deflection for the actual completed mass. Record finished mass and balance.

The owner must then verify exact unidentified electronics connectors, servo voltage/current, control direction, RC failsafe, motor/propeller load and ground clearance. The provisional 522 mm CG mark is not a flight-approved range. No physical load, material forming, aerodynamic or flight test has been claimed.

An independent reviewer checked the electronics guide and flagged the short source nosewire, stale earlier notes and physical fit limits. The revised wire and current guide address those findings; earlier packages and Rev 0 notes are archived outside this ZIP.